Material Characterization Techniques

Lecture given by

Wen Ruitao (温瑞涛)

TA: 李博文

Spring 2021

Evaluation(100):
Class presence/behavior: 10
Mid-term Exam: 40
Final: 50 (exam)

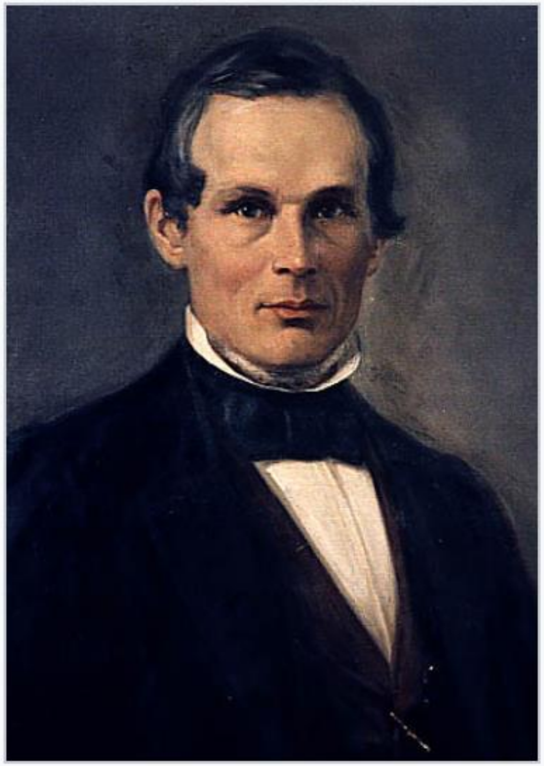
Contents of this course

How small is a nanometer ?

1 nm = 10 Å

= 10^{-9} m, 1 nm = 10^{-3} μm:

roughly the size of 10 hydrogen atoms lined up or the width of a DNA.



Anders Jonas Ångström.

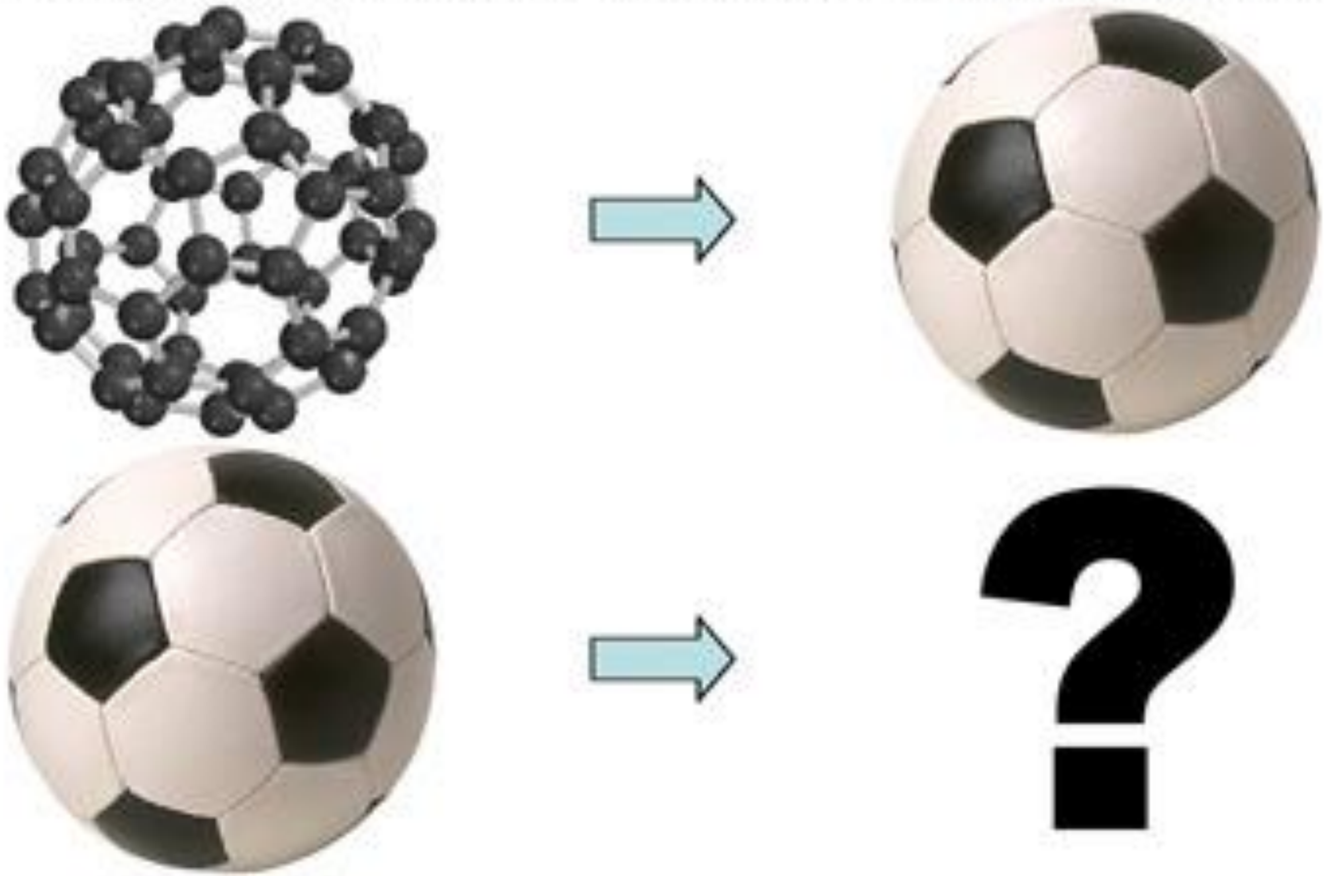
Born	13 August 1814 Lögdö, Medelpad, Sweden
Died	21 June 1874 (aged 59) Uppsala, Sweden
Nationality	Sweden
Citizenship	Swedish
Alma mater	Uppsala University
Known for	Spectroscopy
Awards	Rumford Medal (1872)
	Scientific career
Fields	Physics, Astronomy

- The Ångström **unit (1 Å = 10^{-10} m)** in which the wavelengths of light and interatomic spacings in condensed matter are sometimes measured are named after him.[5] The unit is also used in crystallography as well as spectroscopy.
- The crater Ångström on the Moon is named in his honour.
- One of the main building complexes of Uppsala University, the Ångström Laboratory, is named in his honour. This building houses various departments including the Department of Physics and Astronomy, Department of Mathematics, **Department of Engineering Sciences**, Institute of Space Physics, and the Department of Chemistry.

Scale relation

Nano Fun Facts

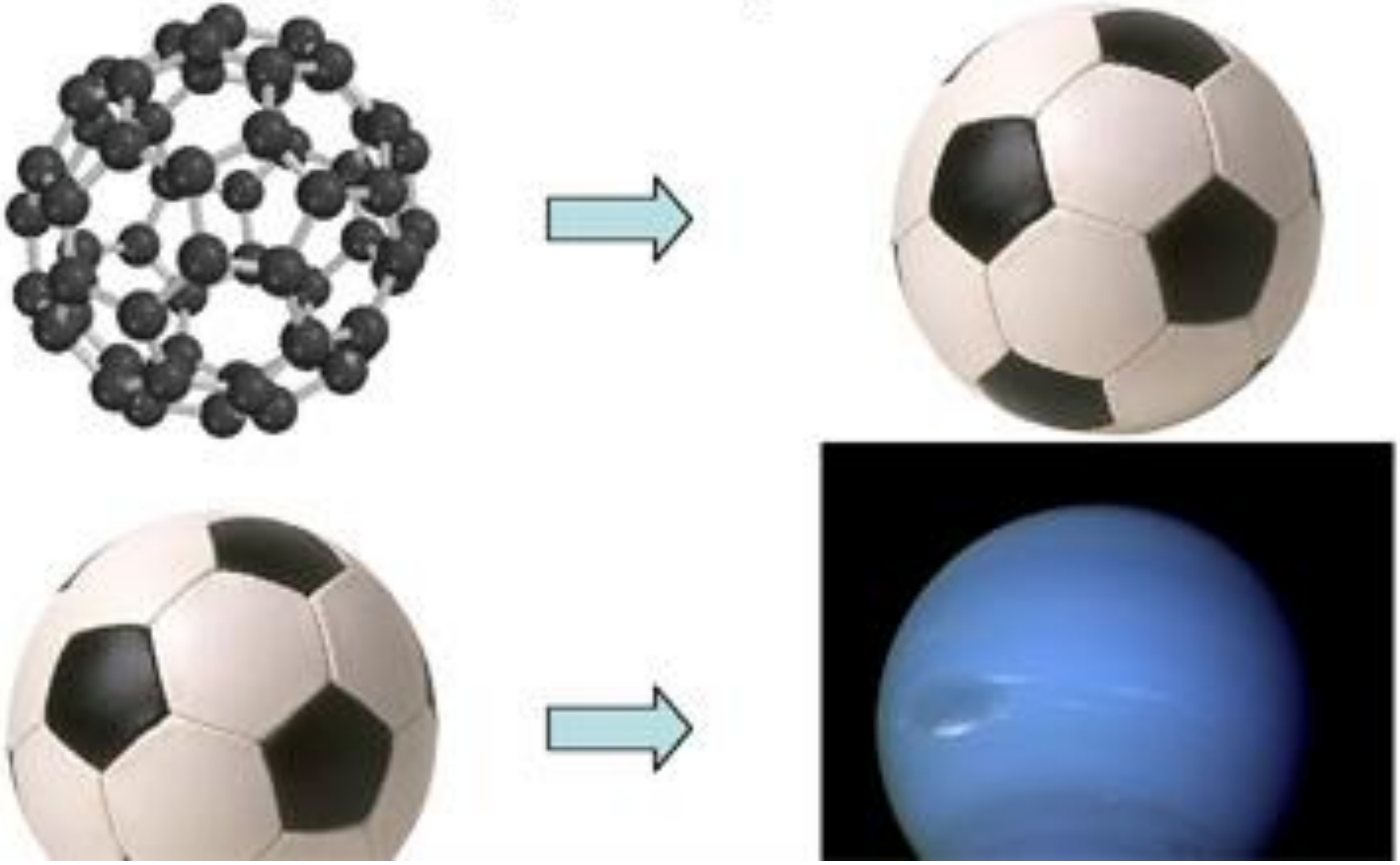
A buckyball is to a soccer ball as a soccer ball is to...



Neptune: 海王星

Nano Fun Facts

A buckyball is to a soccer ball as a soccer ball is to the planet Neptune**(海王星)**



Nano Fun Facts

If the atoms in your body were the size of golf balls, how tall would you be?

Nano Fun Facts

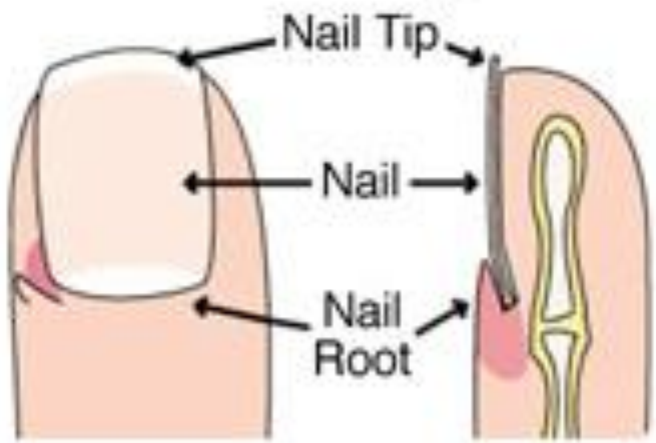
If the atoms in your body were the size of golf balls, how tall would you be?

Nano Fun Facts

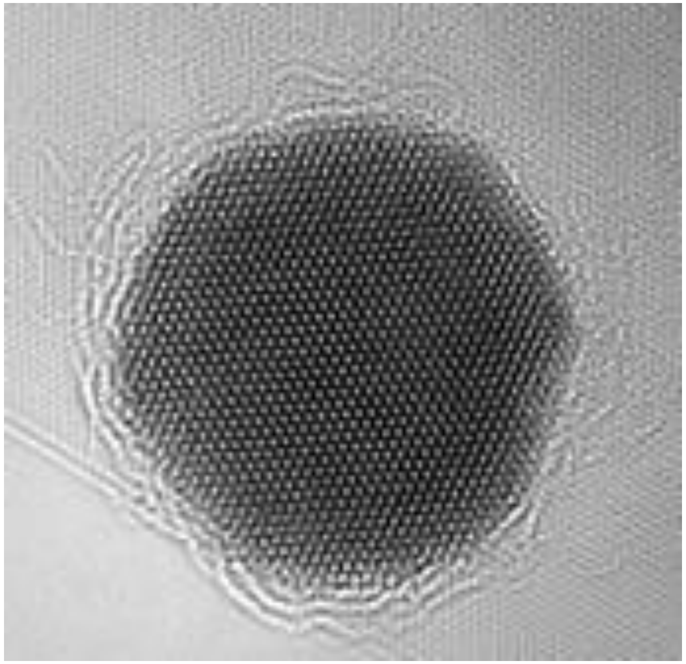
If the atoms in your body were the size of golf balls, you could touch the moon (earth-moon distance = 385000 km).

Nano Fun Facts

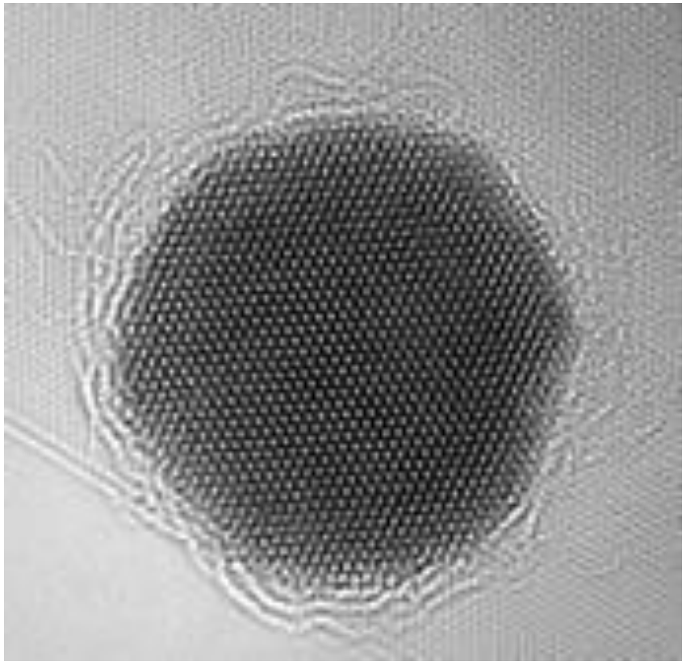
In the time it takes to read this sentence, your fingernails will have grown 1 nm.



Can we see the atom?



Single crystalline (nanoparticle)



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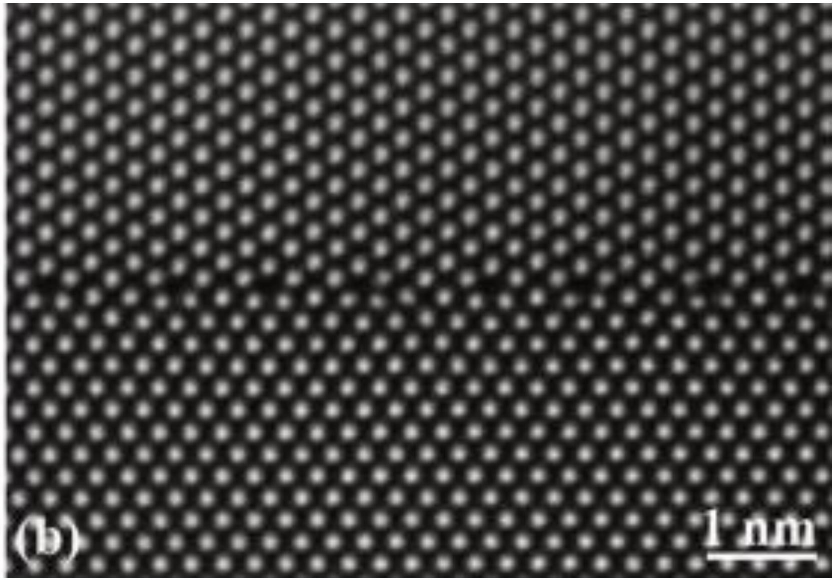
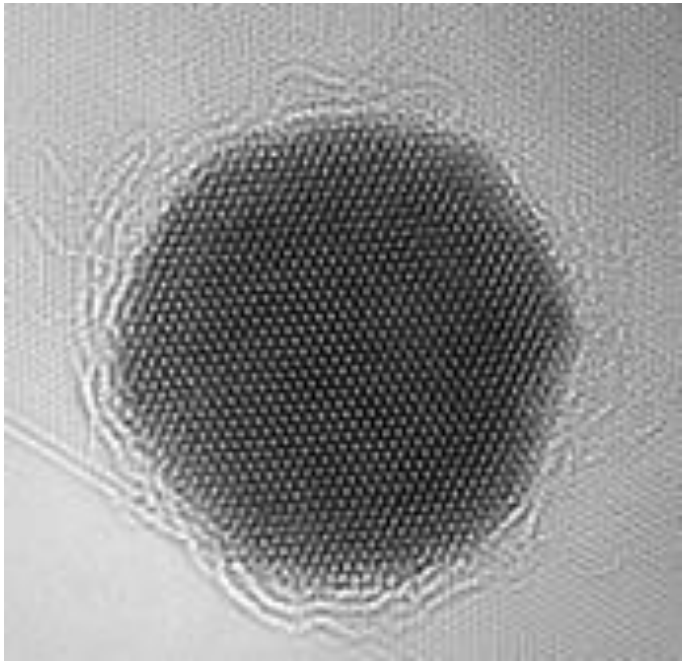


Grain boundary



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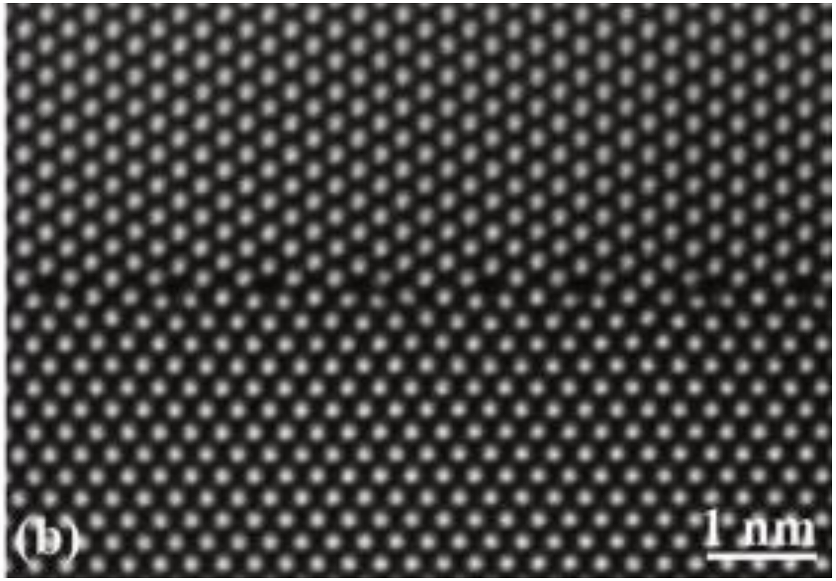
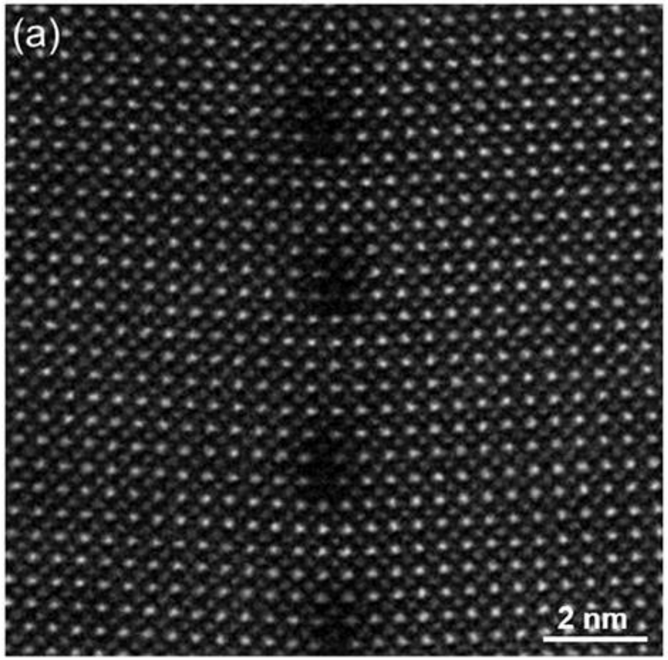
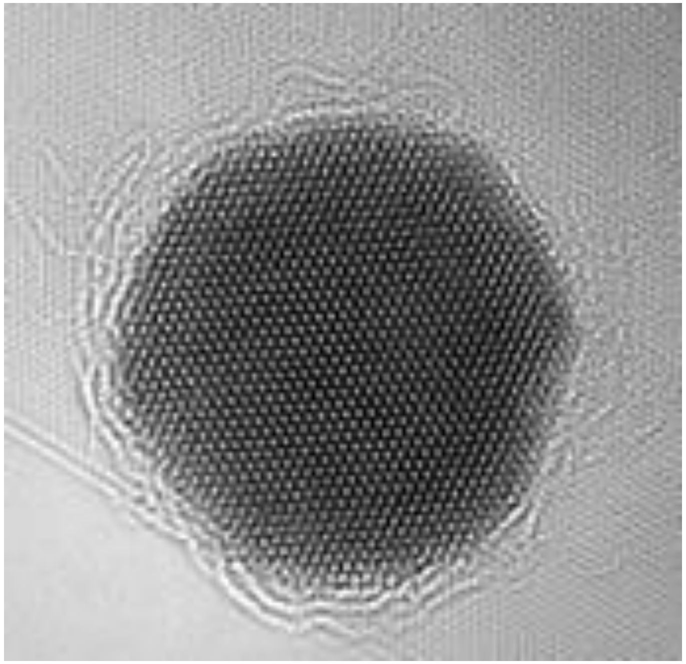


Grain boundary



Dislocations



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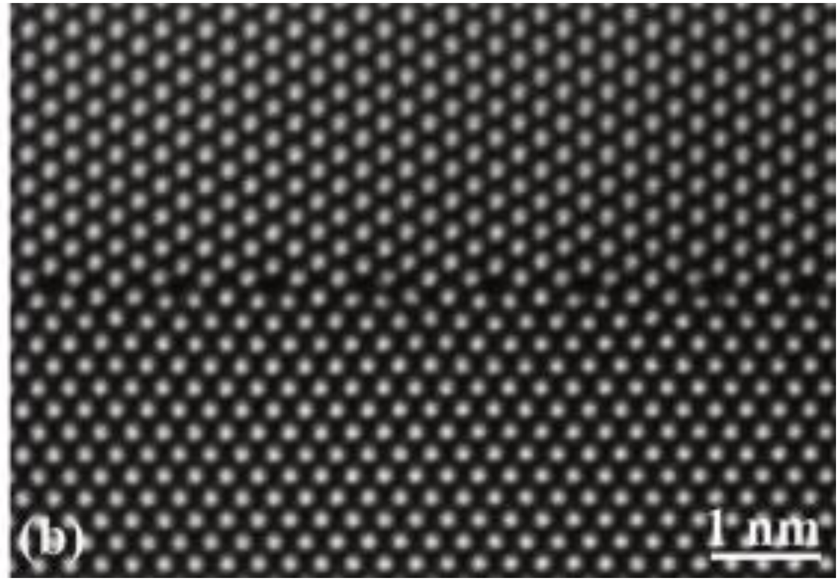
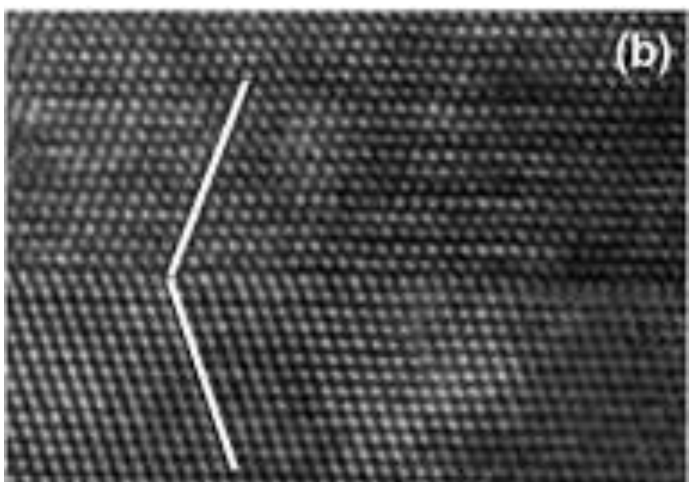
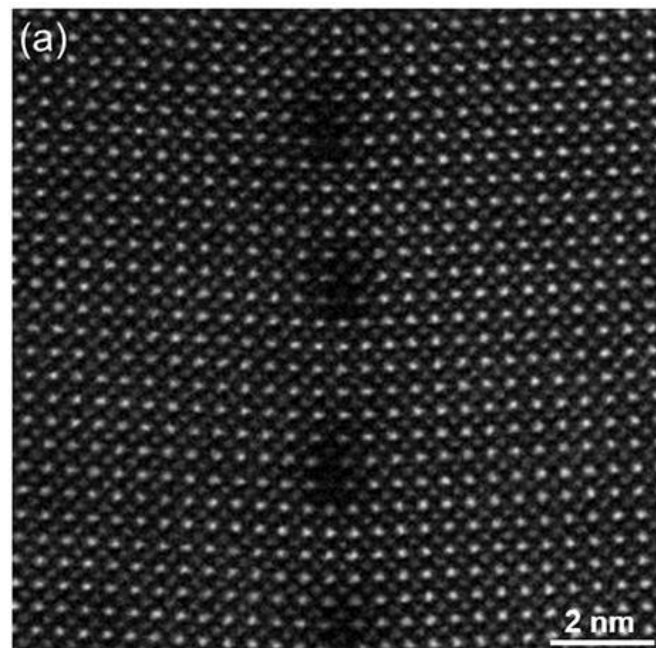


Grain boundary



Twin



Dislocations

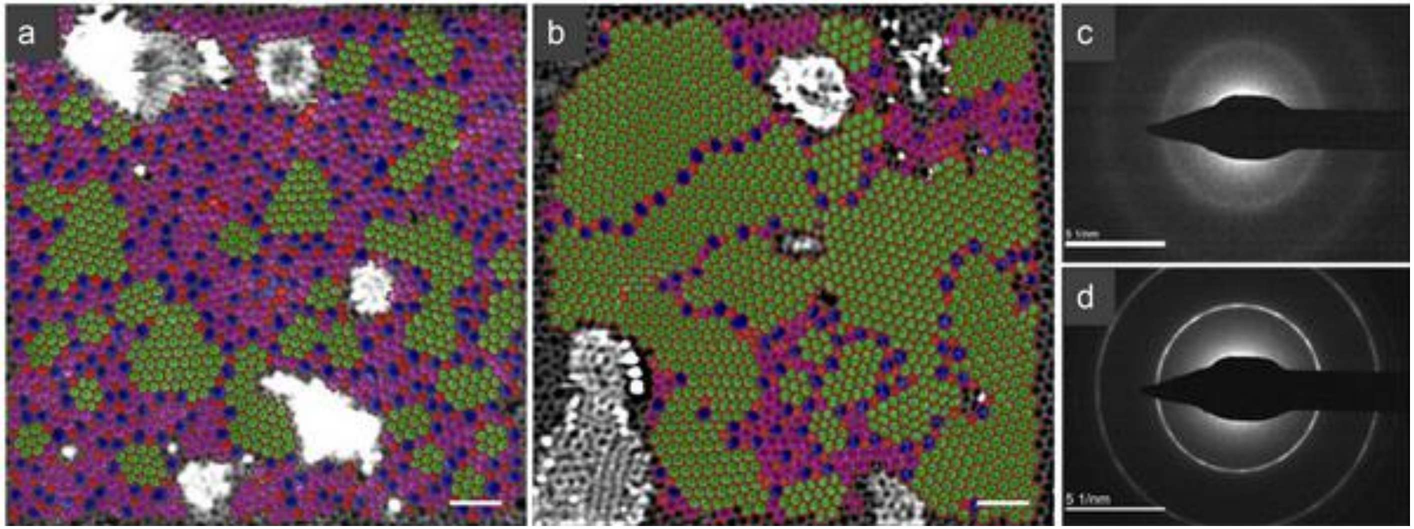
Article | Published: 08 January 2020

Synthesis and properties of free-standing monolayer amorphous carbon

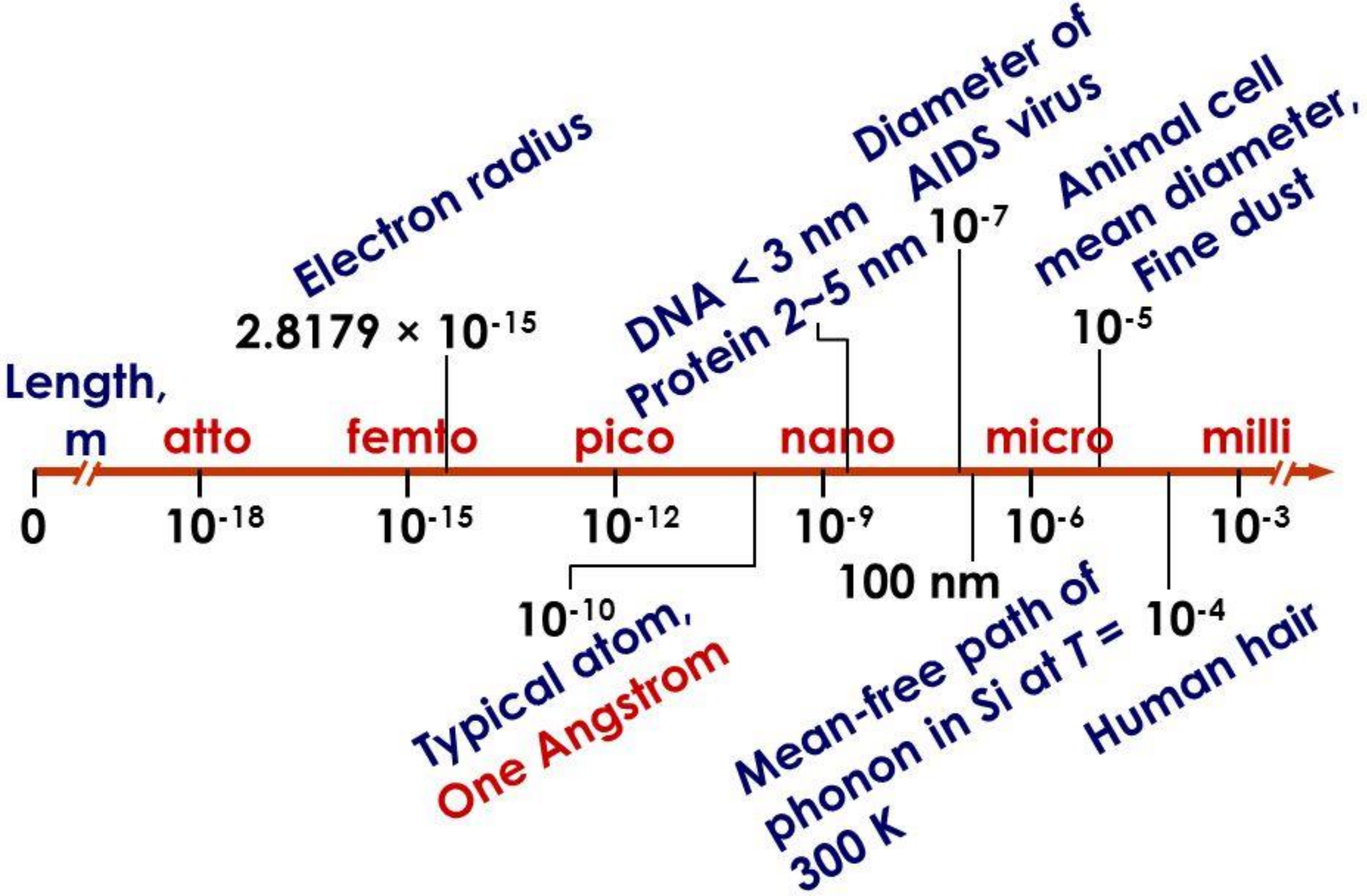
Chee-Tat Toh, Hongji Zhang, Junhao Lin, Alexander S. Mayorov, Yun-Peng Wang, Carlo M. Orofeo, Darim Badur Ferry, Henrik Andersen, Nurbek Kakenov, Zenglong Guo, Irfan Haider Abidi, Hunter Sims, Kazu Suenaga, Sokrates T. Pantelides & Barbaros Özyilmaz

Nature **577**, 199–203(2020) | Cite this article

Amorphous carbon atoms

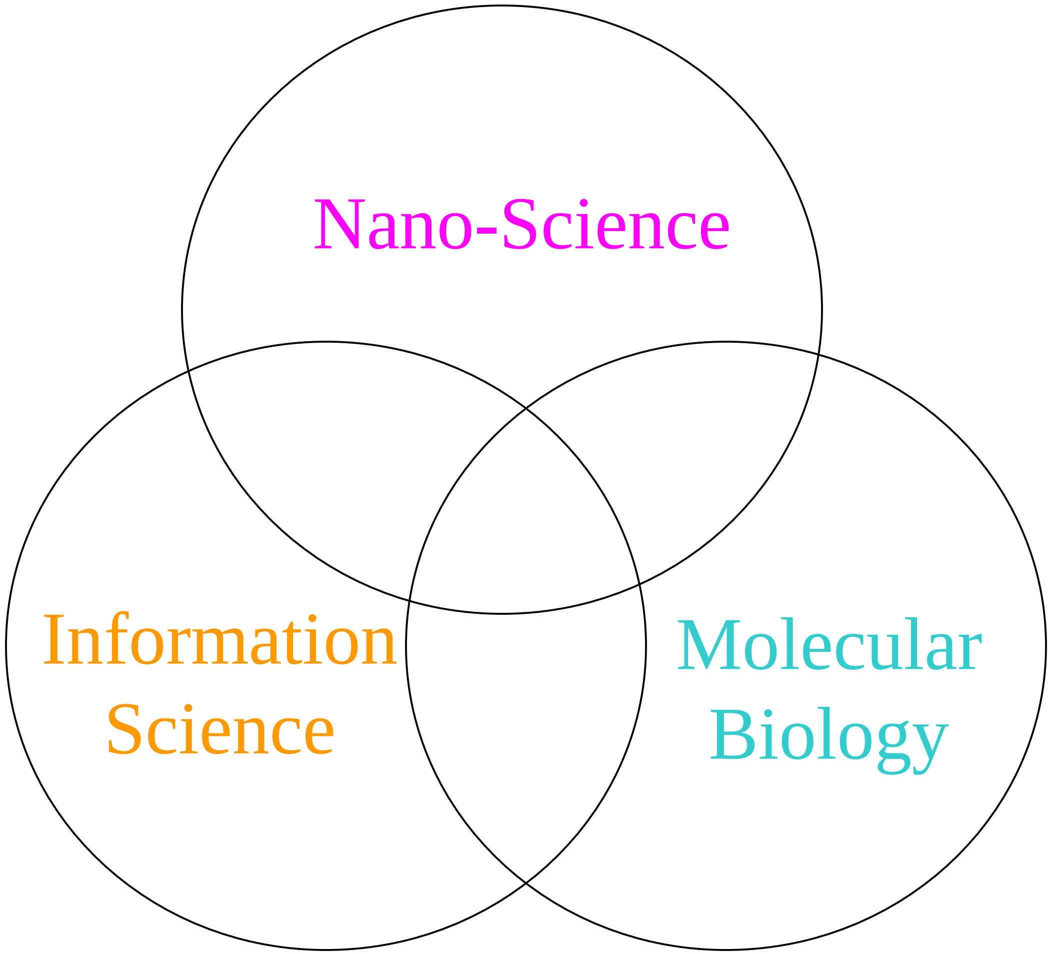


Length Scales



- Optical microscope
- Scanning electron microscope (SEM)
- Transmission electron microscope (TEM)
- Atom force microscope (AFM)
- Scanning tunneling microscope (STM)
- etc.

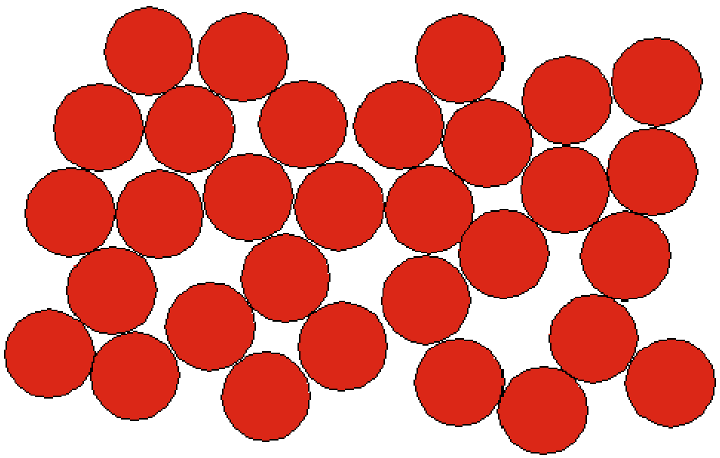
Major Scientific Disciplines in the New Millenium (千禧年)

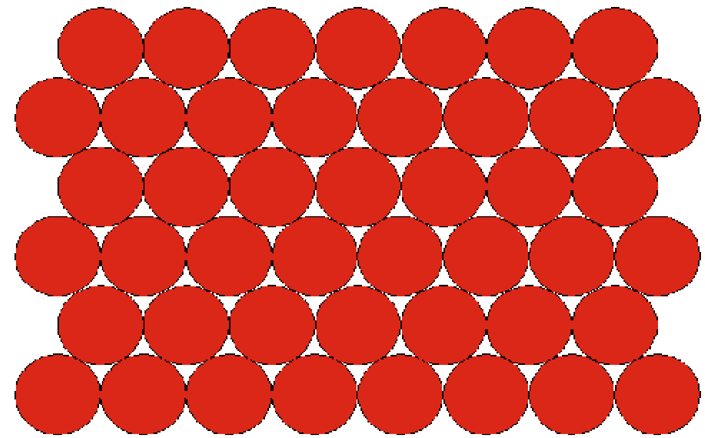


What is nanoscience?
a science of understanding the tiny.

What is nanotechnology?
the controlled manipulation of matter at the nanometre length scale

Close Packing Concept

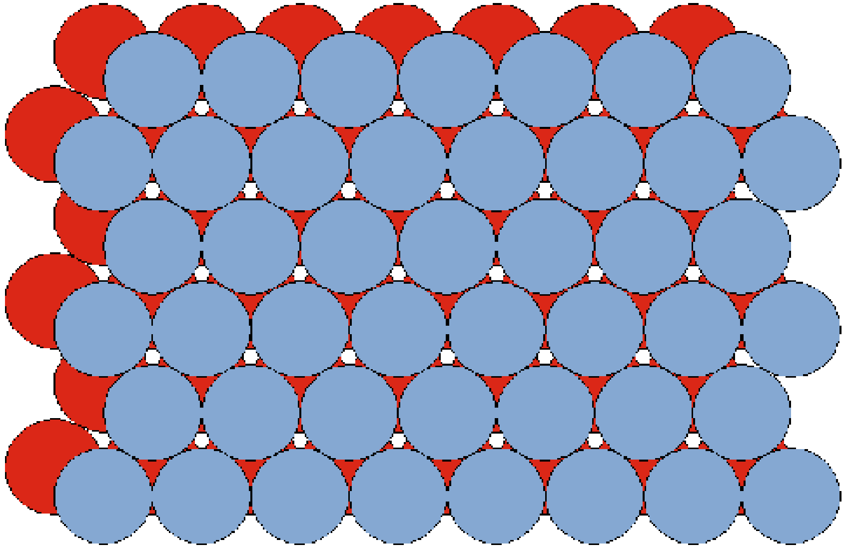




- Random arrangement of atoms (hard, neutral spheres)

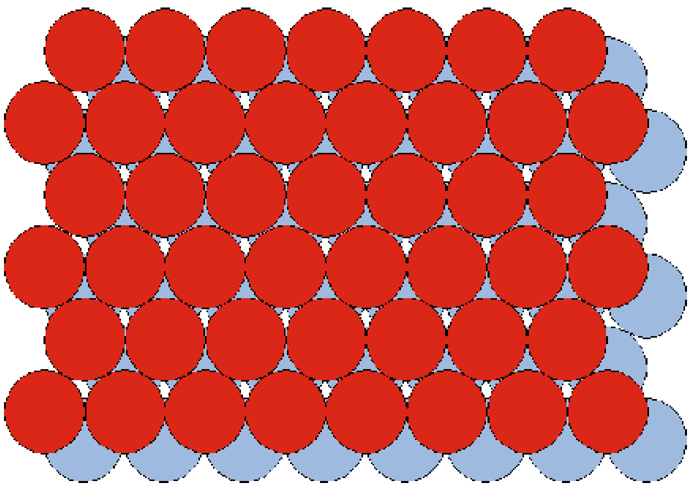
- 2D close-packed layer
- Most efficient way to fill space

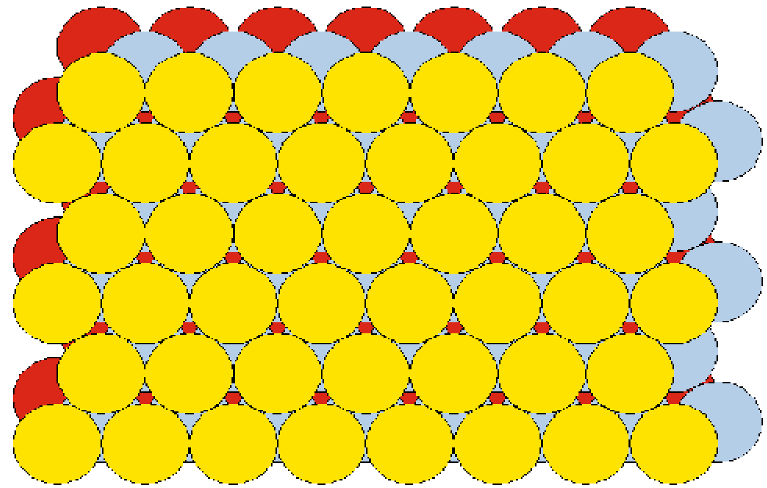
“Building” a Close Packed Lattice



- Second identical (grey) layer of close-packed spheres
- Staggered onto first (red) layer

Cubic and Hexagonal Close Packed Lattices

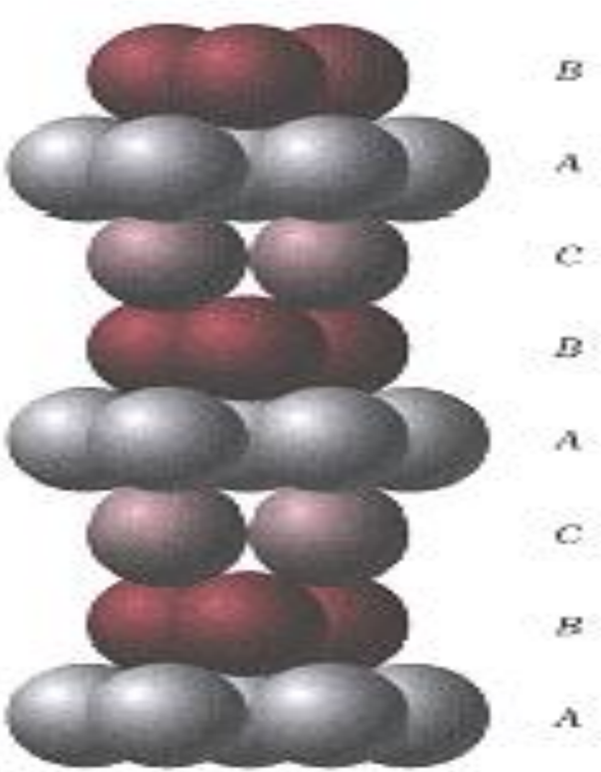


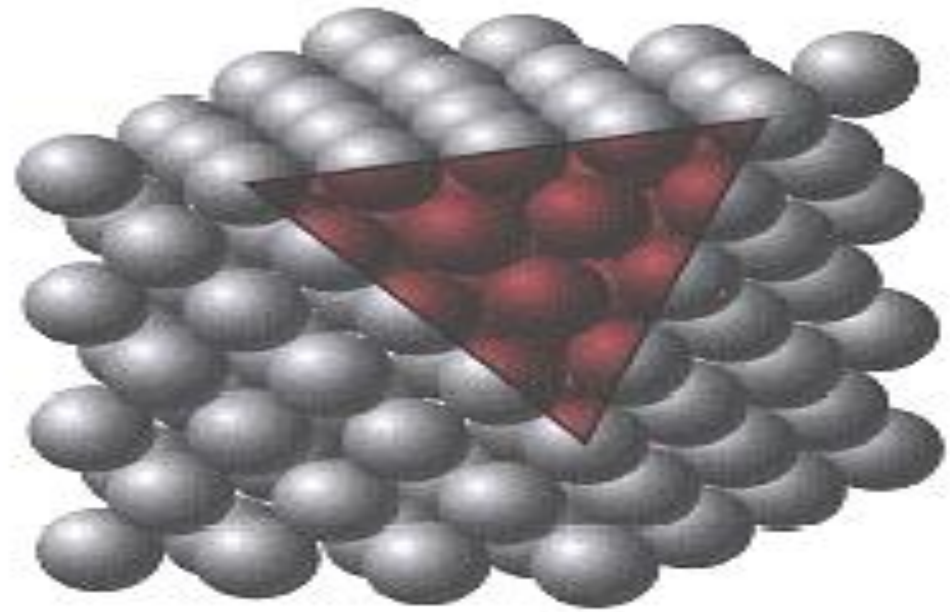


- Third (red) layer above first (red) layer
- ABABAB ≡ hcp

- Third (yellow) layer not above first or second layer
- ABCABC ≡ ccp ≡ fcc

FCC: Stacking Sequence ABCABCABC...

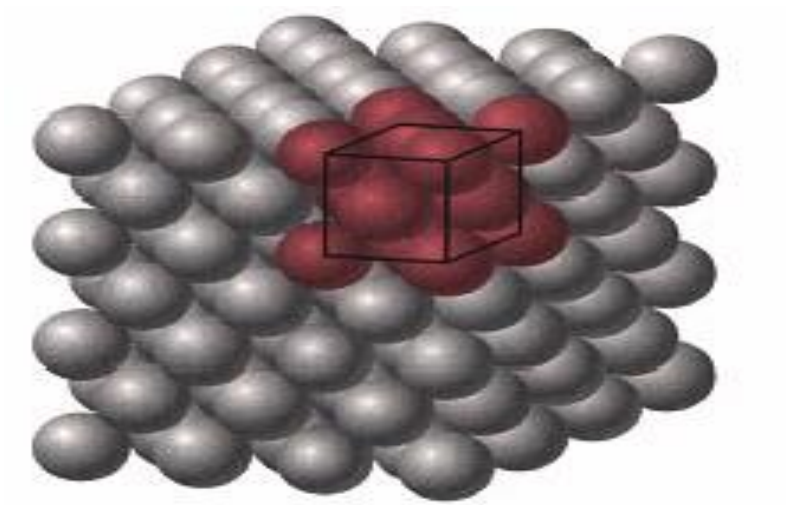


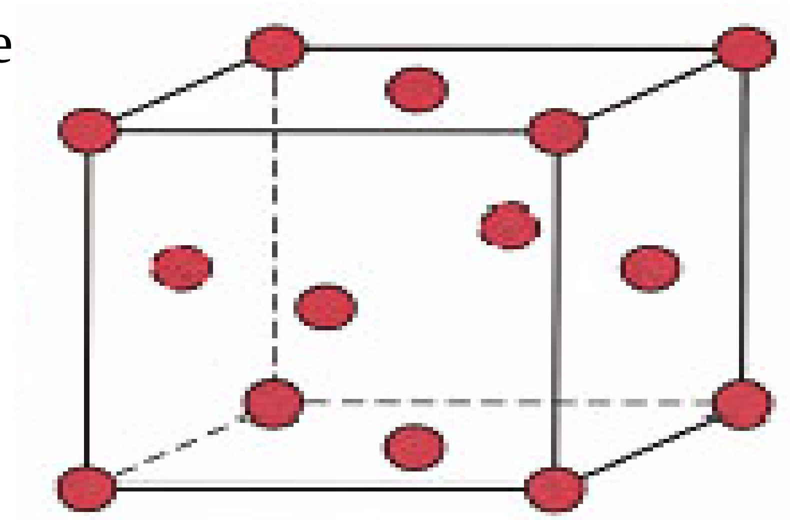


Third plane is placed above the “holes” of the first plane not covered by the second plane

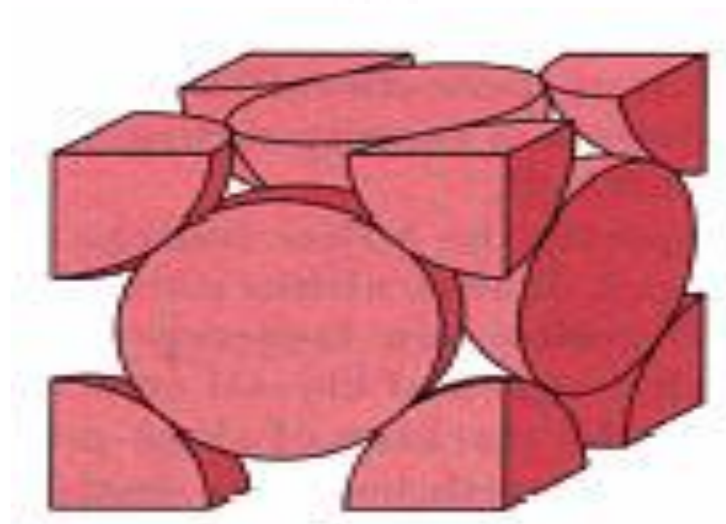
Face-Centered Cubic (FCC) Crystal Structure (I)

- Atoms are located at each of the corners and on the centers of all the faces of cubic unit cell
- Cu, Al, Ag, Au have this crystal structure

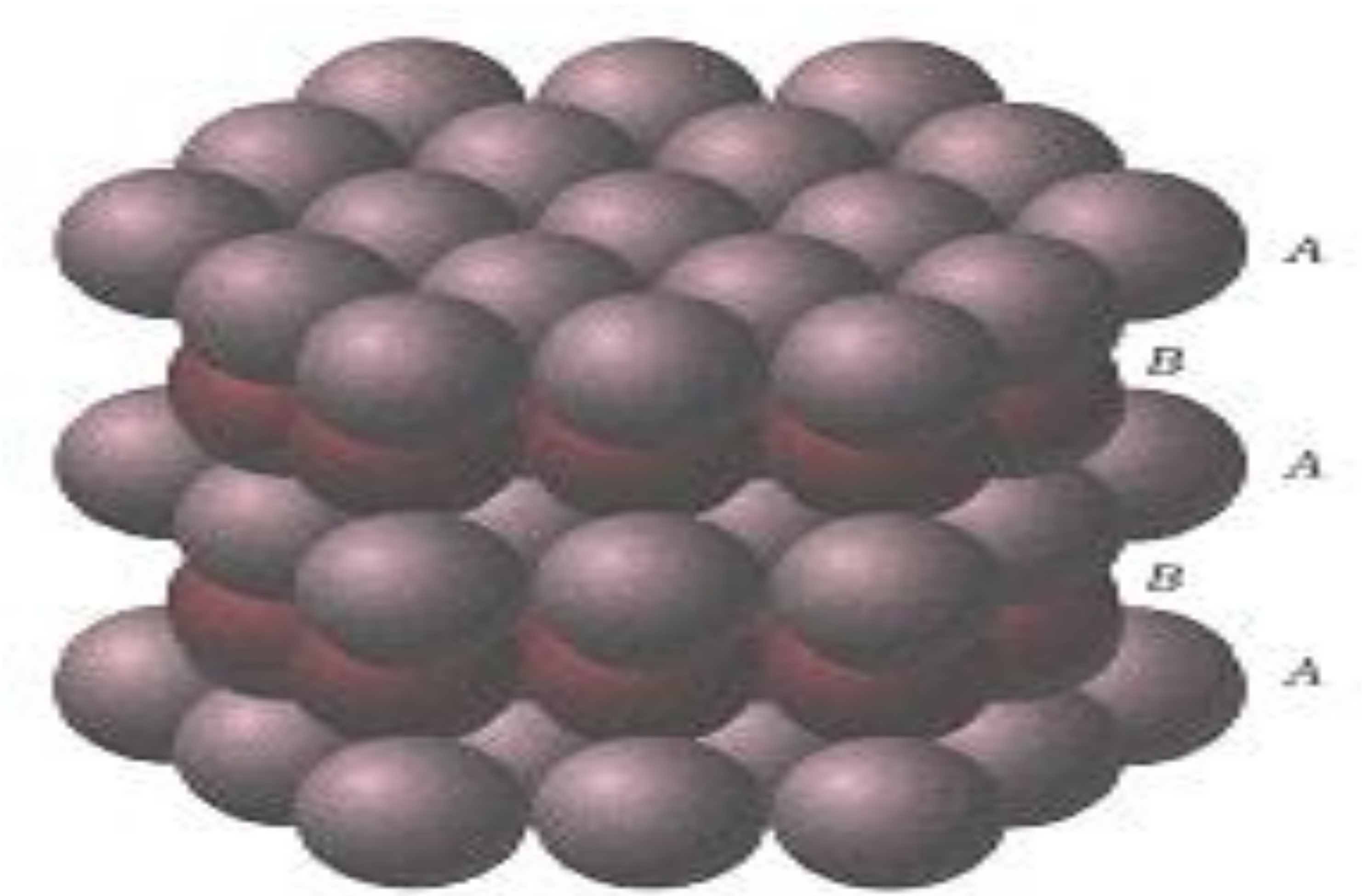




Two representations of the FCC unit cell



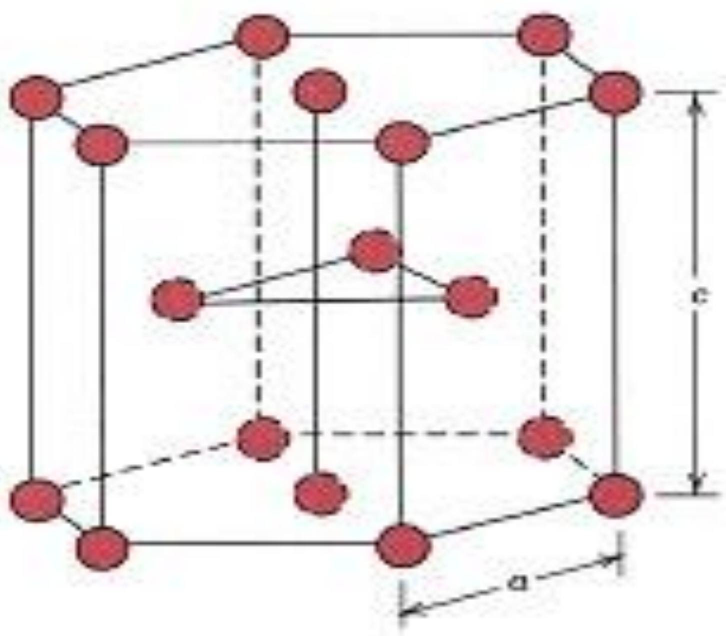
HCP: Stacking Sequence ABABAB...

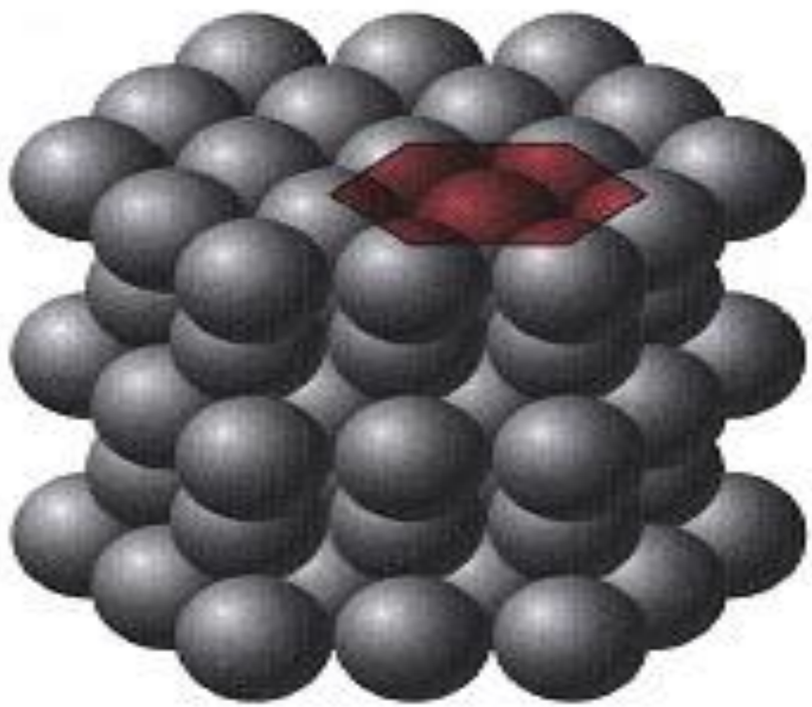


Third plane is placed directly above the first plane of atoms

Hexagonal Close-Packed Crystal Structure (I)

- HCP is one more common structure of metallic crystals
- Six atoms form regular hexagon, surrounding one atom in center. Another plane is situated halfway up unit cell (c-axis), with 3 additional atoms situated at interstices of hexagonal (close-packed) planes
- Mg, Zn, Ti have this crystal structure





Formation of a crystalline structure is in a natural way.

Can we manipulate the atoms?
Or can we move the atoms and stack them at our will?

“There's Plenty of Room at the Bottom”

An Invitation to Enter a New Field of Physics

Richard Feynman 29/12/1959

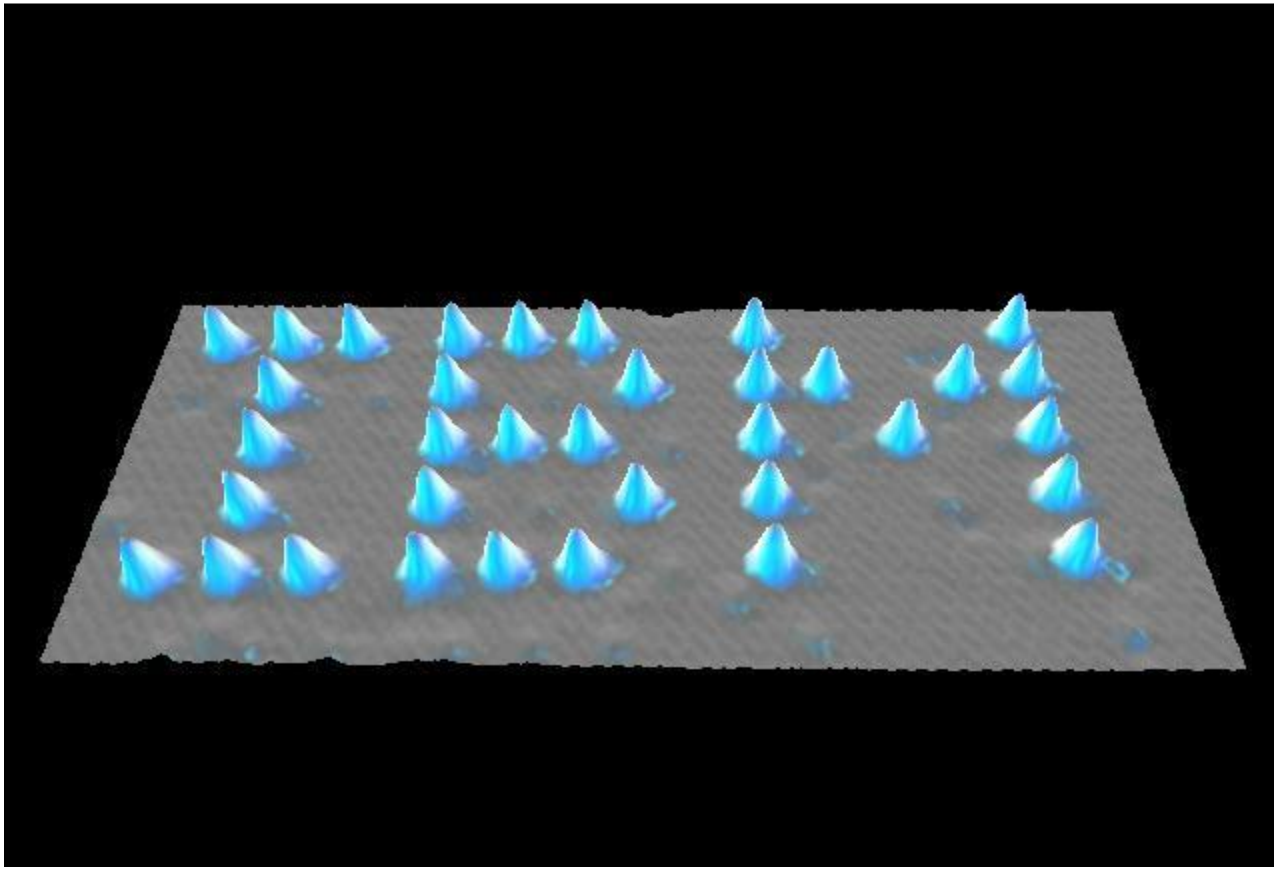
“I am not afraid to consider the final question as to whether, --we can arrange the atoms the way we want; the very atoms, all the way down!

The principles of physics do not speak against the possibility of maneuvering things atom by atom.

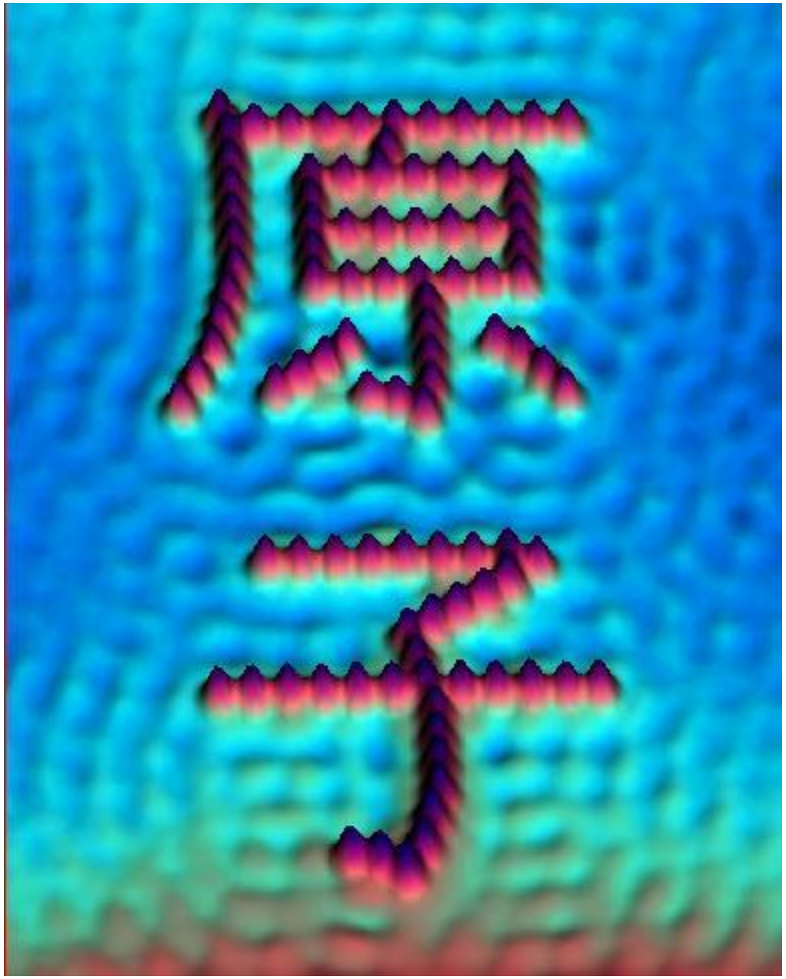
What would happen if we could arrange the atoms one by one the way we want them …”

Single Atom Manipulation
Xenon on Nickel (110)

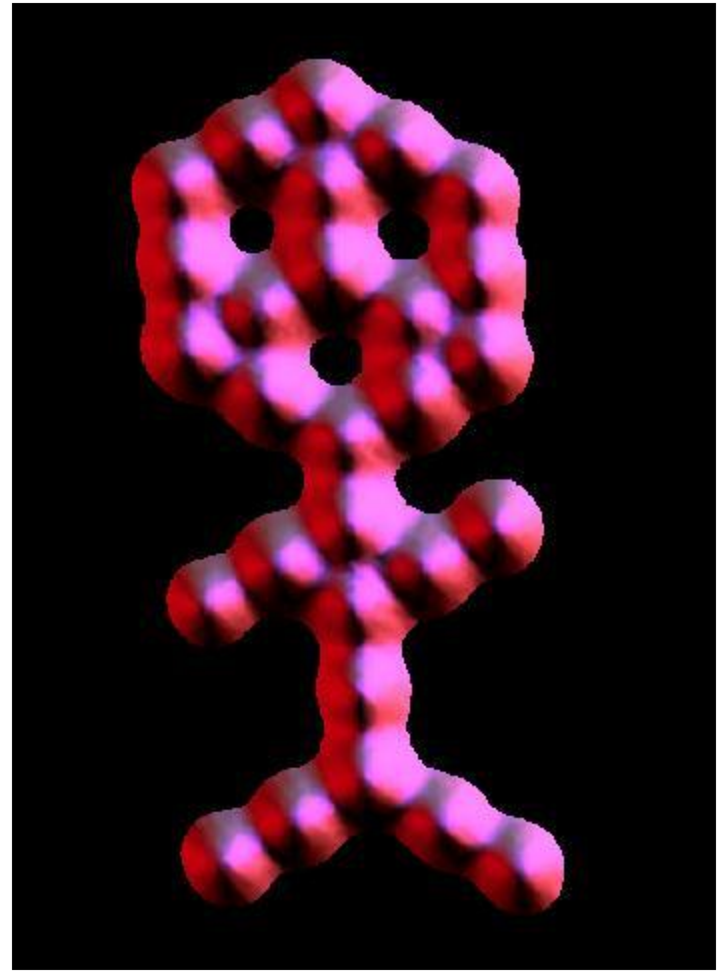
Xenon:
氙



Ref: D.M. Eigler, E.K. Schweizer. Positioning single atoms with a scanning tunneling microscope. Nature 344, 524-526 (1990).



“Atom”
Iron on Copper (111)



“Carbon Monoxide Man”
CO on Platinum (111)

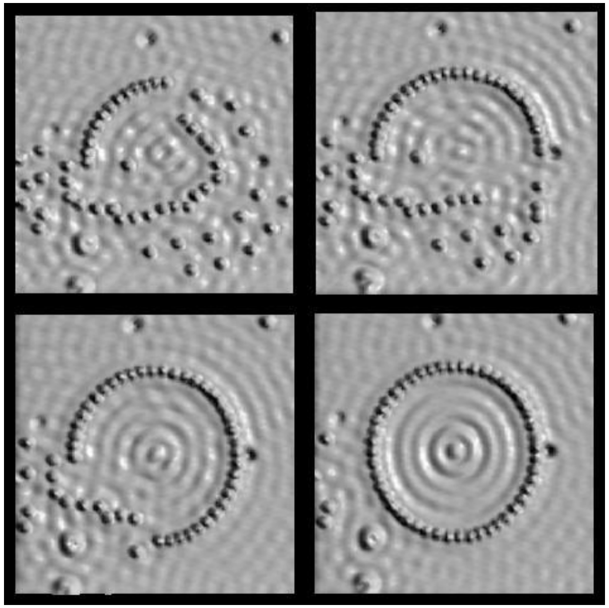
Introduction: Philosophy

- Old philosophy of creating things was to start with something big, and make it smaller.
- Nanoscience starts with something atomic and builds things with it.

The difference between Micro and Nano

"In microtechnology, the challenge is to build smaller"
• starting from macroscopic material and reducing the size,
e.g. lithography, MEMS etc.

"In nanotechnology, the challenge is to build bigger"
• building up from individual atoms

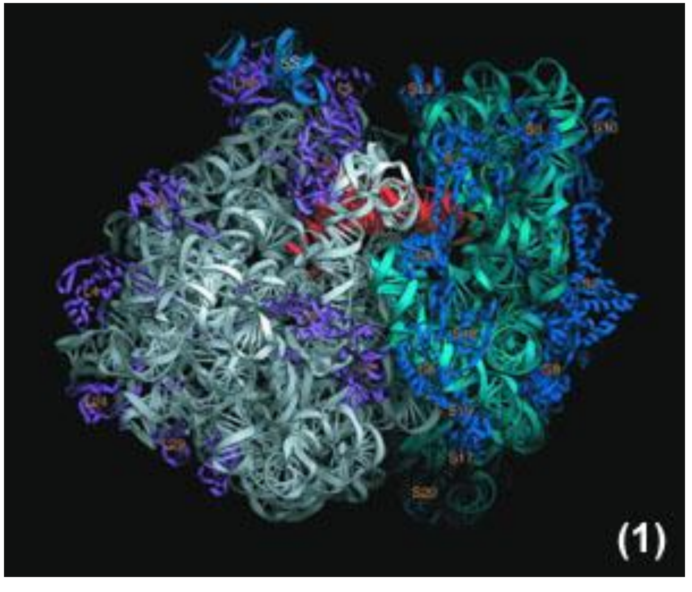


The Making of the Circular Corral
Iron on Copper (111)

DNA includes genetic information of an organism. The same DNA can be arranged in different sequences. As a result, different organsms can be generated (e.g., human, monkey).

This level of information compression would be considered an amazing engineering achievement, if it had been created by humankind.

Proteins and enzymes(蛋白质和酶):
These large polymers (or polymer aggregates) adopt very specific, unique conformations(构造) in solution.
The specific shape of a protein is responsible for activity.

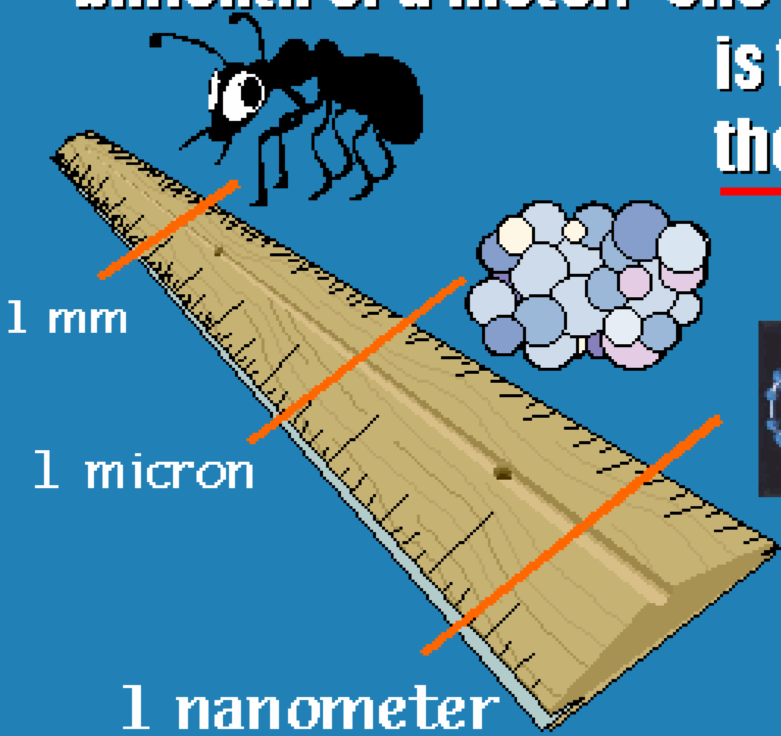


(1)

It is here that the remarkable specificity of biological chemistry begins.

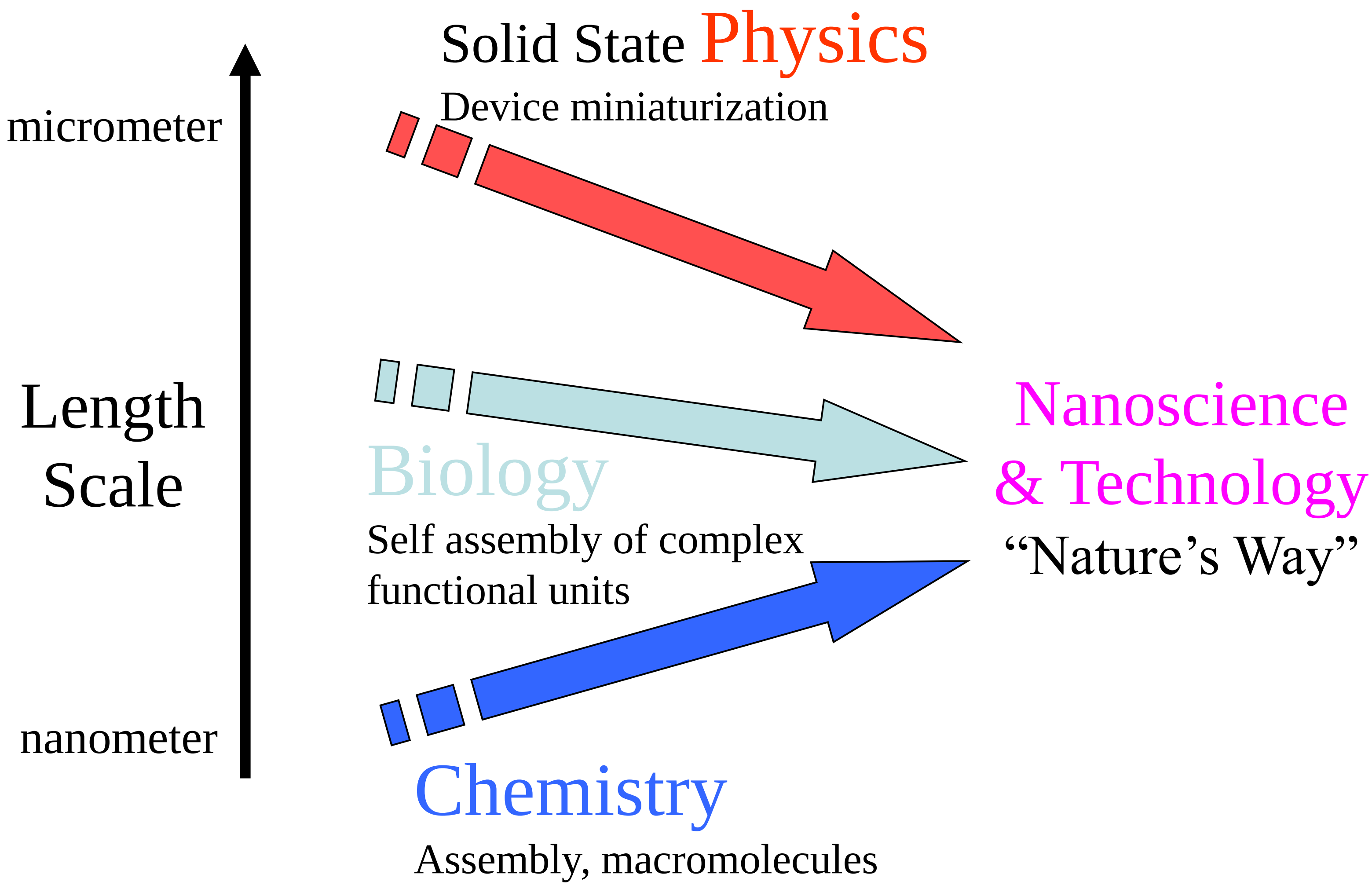
Definition of Nanoscience

The word "nano" means 10^{-9} so a nanometer is one billionth of a meter. One definition of nanoscience is that it concerns itself with the study of objects which are anywhere from hundreds to one of nanometers in size.



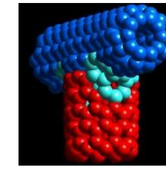
1 – 1000 nm

Integration of traditional disciplines



Nanoscience is an emerging area of science which concerns itself with the study of materials that have very, very small dimensions.
It can’t really be called chemistry, physics or biology; All sorts of scientists are studying very small things in order to better understand our world.

What is Special about Nanoscale?



Condensed matter physics: solids

Nanoscience

Chemistry: atoms and molecules

Quantum chemistry does not apply (although fundamental laws hold) and the systems are not large enough for classical laws of physics

Atoms and molecules are generally less than 1 nm and we study them in chemistry.

Condensed matter physics deals with solids with infinite array of bound atoms.

Nanoscience deals with the in-between meso-world

x

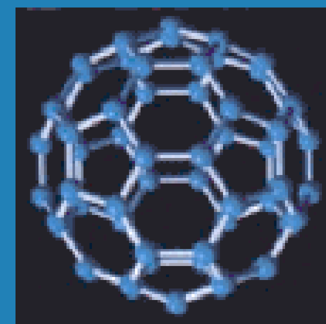
- Size-dependent properties

- Surface to volume ratio
  - A 3 nm Fe particle has 50% atoms on the surface
  - A 10 nm Fe particle has 20% atoms on the surface
  - A 30 nm Fe particle has only 5% atoms on the surface

What is Special about Nanoscale?

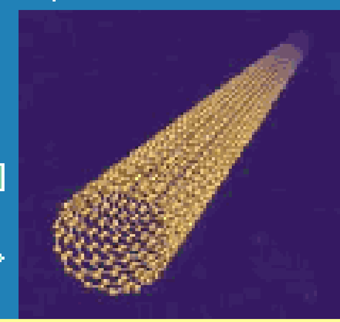
Small Solids Look Different

- Nanoscience would be boring if small things were just like big things. Luckily they are not. Pencil lead, graphite for example takes on all sorts of interesting shapes if it is kept from becoming a big solid.



C_{60} The Buckyball

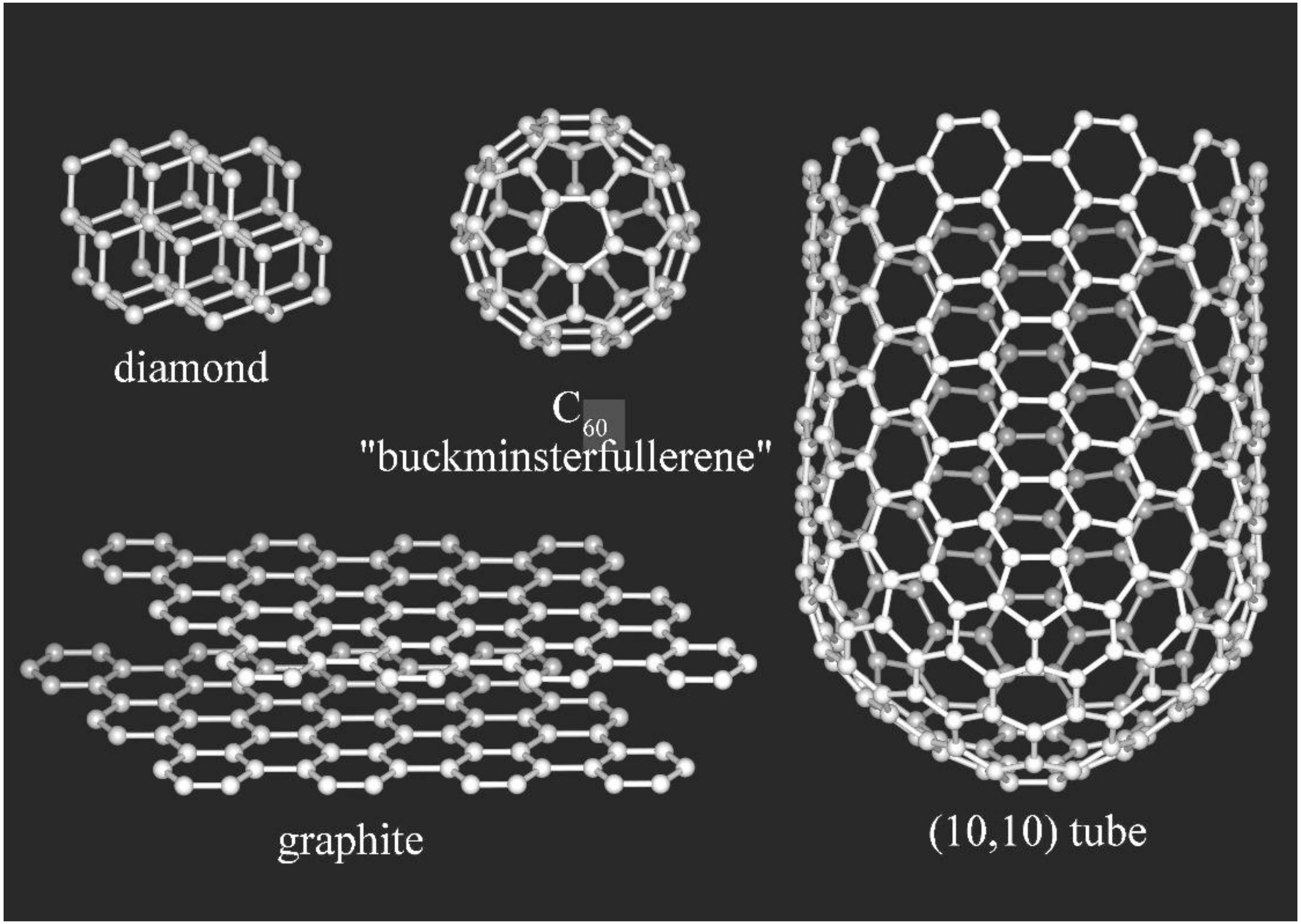
A Carbon Nanotube



The Many Shapes of Carbon

When carbon is a pure solid it typically is found as graphite or even rarer as diamond. However, on the nanoscale carbon takes on very different structures! Rick Smalley, Bob Curl and Harry Kroto won the 1996 Nobel Prize in Chemistry for the discovery of C60. Check out Dr. Smalley's web page for the latest in carbon nanoscience!

Carbon Atoms

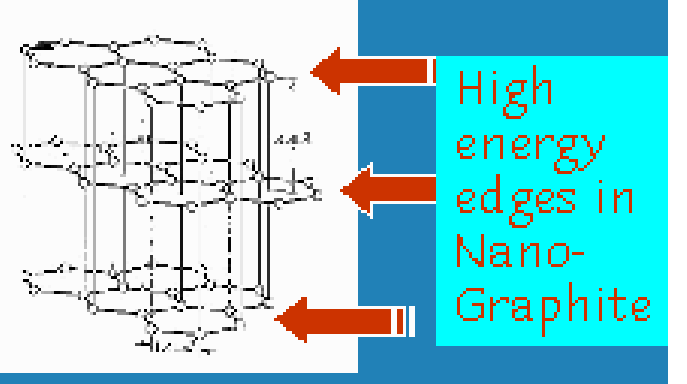


Different materials can be produced even from the same atoms, but by different arrangements.

X

Edges and Surfaces Matter

The reason so many nanosolids look different than their larger cousins is primarily due to their edges. A tiny piece of graphite would have many of its atoms at its edges, which are unstable. Most nanosolids are metastable: if given an opportunity they would combine to become larger.



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Counting the Edges of Nanosolids

- In the case of carbon, nanosolids will roll themselves into shapes that have few, if any edges! Big carbon pieces, like in your pencil, also have edges but they make up a very small fraction of all of the atoms.

Nanotubes:
Rolled Sheets of
Graphite with no Edges

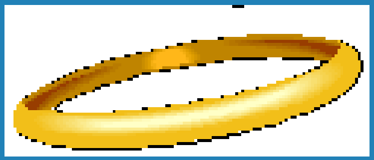
What Shape Will it Be?

It is important that nanoscientists be able to predict the shapes of their nanosolids. Theoretical chemistry can calculate precisely the energy costs associated with carbon edges as well as sheet curvatures. Then, just like balancing a checkbook, the total energy of various structures can be found. See Gus Scuseria's web page for more information!

What is Special about Nanoscale?

Small Solids Act Different

- Not only are the shapes and structures of many nanosolids different from the bulk, but so are their properties. You think of the metal gold as a yellowish solid; a collection of nanometer sized pieces of gold might look more reddish.

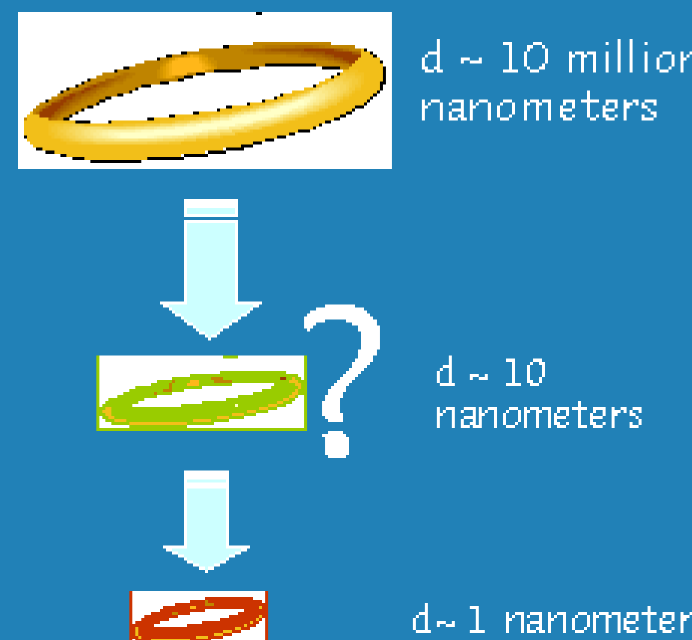


Bulk Gold= Yellow

Nanogold = Red

The Color of Small Solids

- If nanosolids were just different shapes or colors, then studying them would be just a curiosity. The real power lies in our ability to ask: “How does the property change with the size?”



Nanoscience and Nanotechnology

- The nanoscale is not just another step towards miniaturization (小型化). It is a qualitatively new scale where materials properties, such as melting point or electrical conductivity, differ significantly from the same properties in the bulk.
  - “Nanoscience” seeks to understand these new properties.
  - “Nanotechnology” seeks to develop materials and structures that exhibit novel and significantly improved physical, chemical, and biological properties and functions due to their nanoscale size.

The goals of nanoscience and nanotechnology are:

to understand and predict the properties of materials at the nanoscale

to "manufacture" nanoscale components from the bottom up

to integrate nanoscale components into macroscopic scale objects and devices for real-world uses

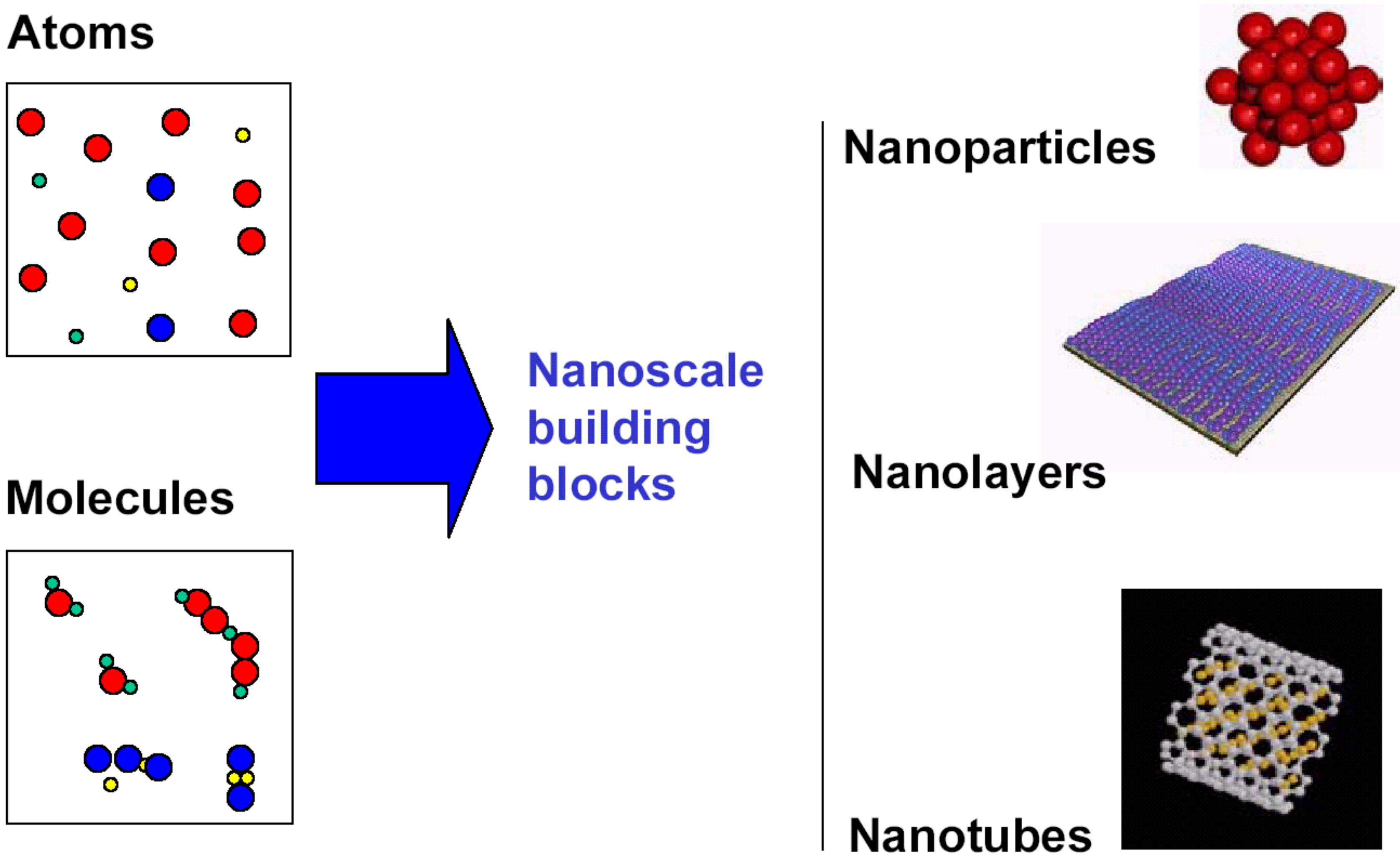
Nanoscience and Nanotechnology

- Nanomaterials
- Nanofabrication
- Nanoelectronics
- Nanobiology/nanomedicine

Nanomaterials

materials are needed to make anything in technology…

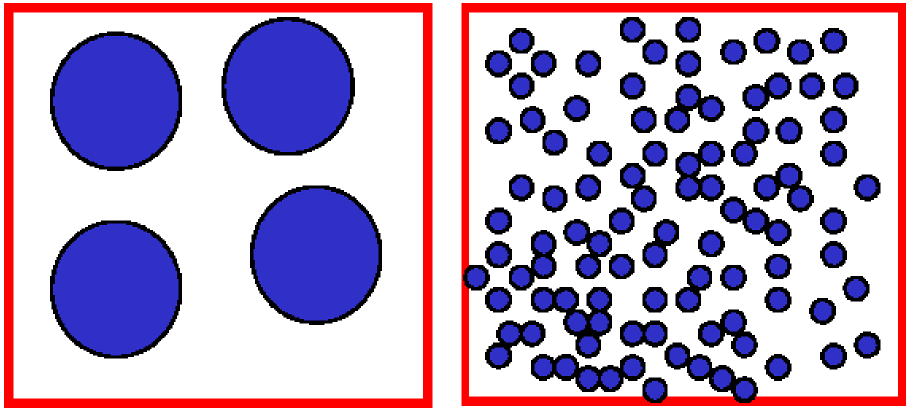
nanomaterials are needed to make anything in nanotechnology


Atoms
Molecules
Nanoscale
building
blocks
Nanoparticles
Nanolayers
Nanotubes

What is special about nanoscale building blocks?

- Size confinement

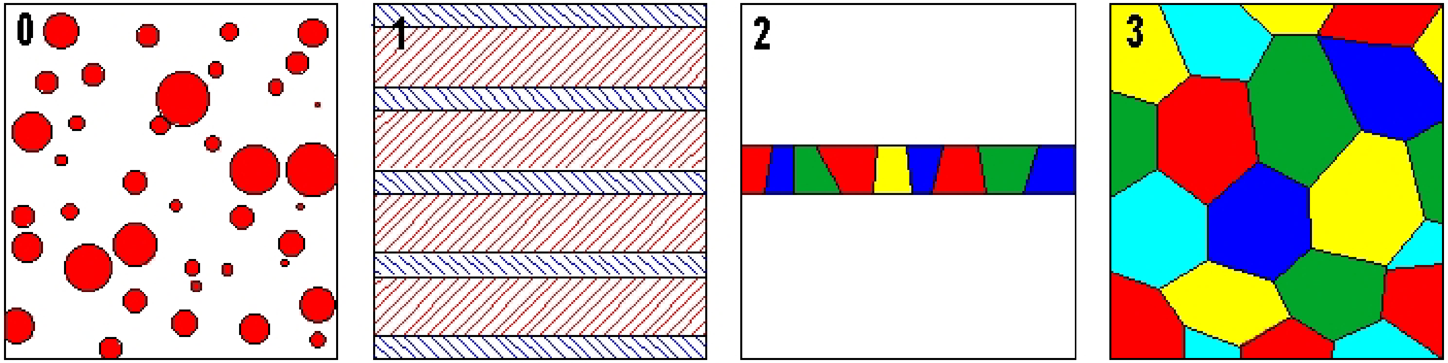
- High surface area
- Many interfaces



What are nanostructured materials?

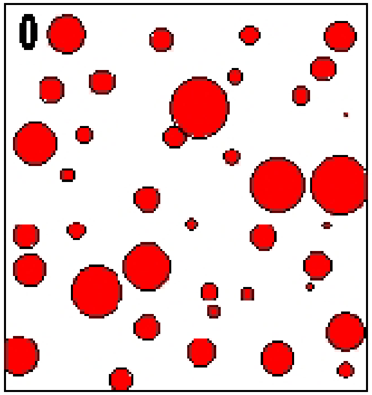
- Any material with at least one dimension in the 1-100 nm range.

Classes of nanostructured materials:



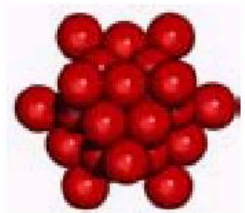
There are several different types of nanostructured materials. These range from zero dimensional atom clusters to three dimensional equi-axed grain structure. Each class has at least one dimension in the nanometer range, as shown in the figure.

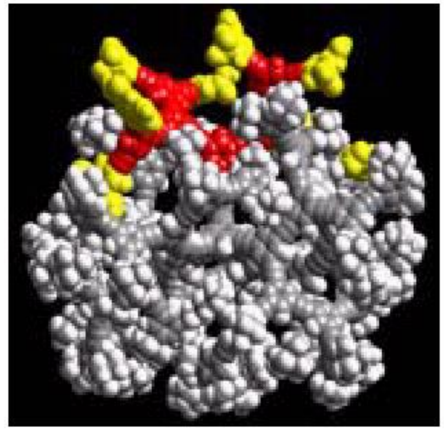
(from R.W. Siegel, Nanophase Materials, Encyclopedia of Applied Physics, vol. 11, VCH Publishers 1994, p 173)

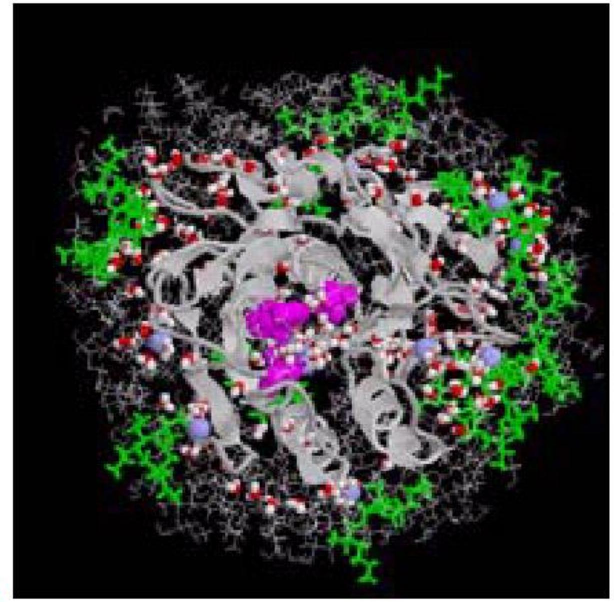


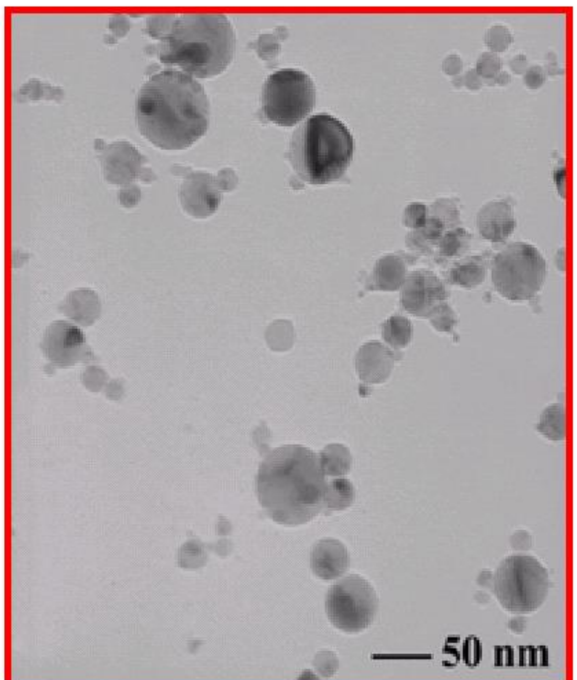
Atom clusters or particles with sizes in the nanometer range.

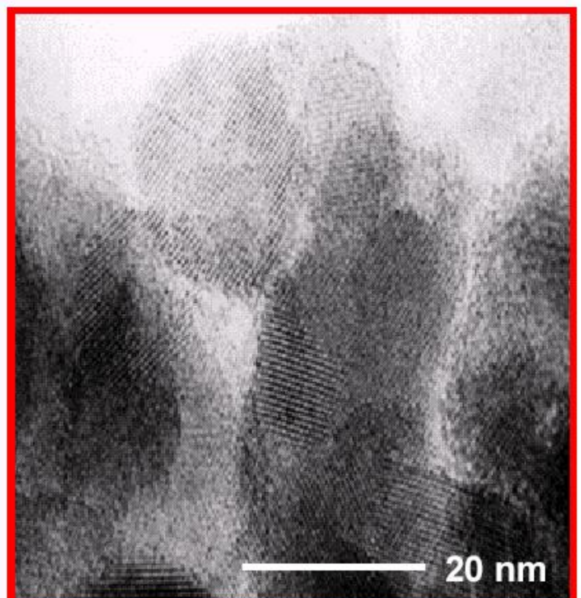
Nanoparticles

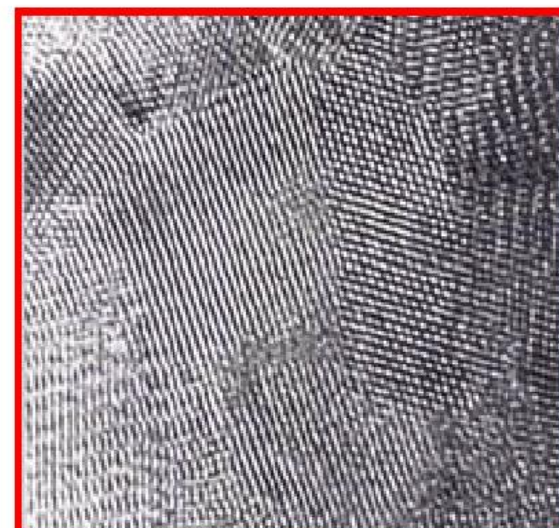






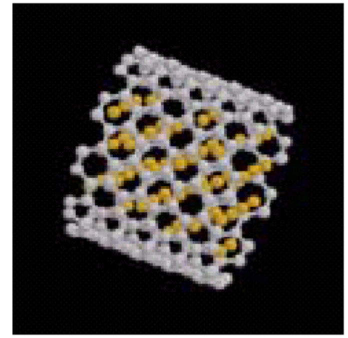


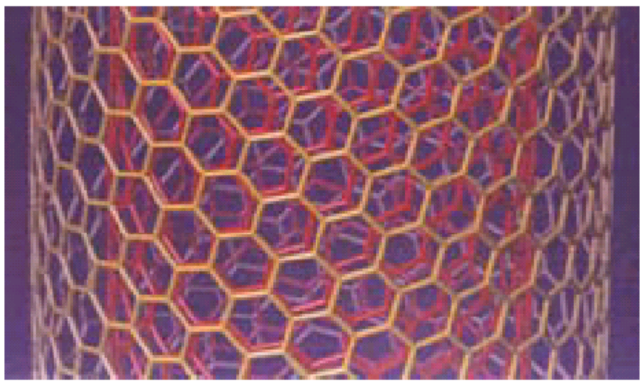


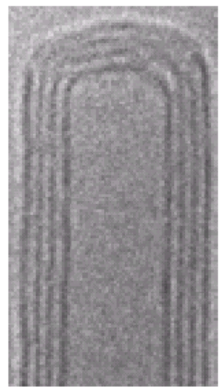


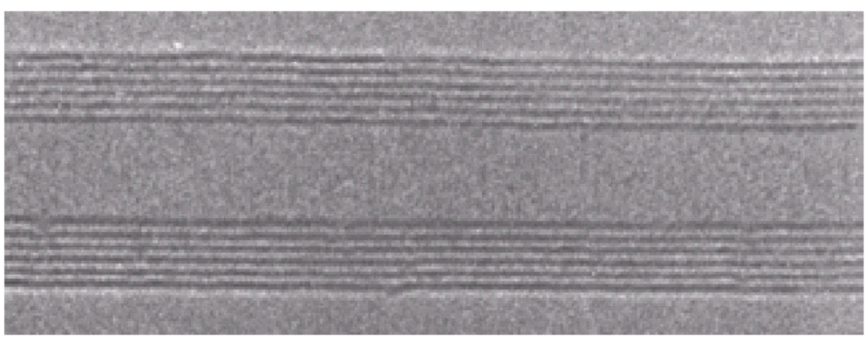
Any tubes with diameters in the nanometer range.

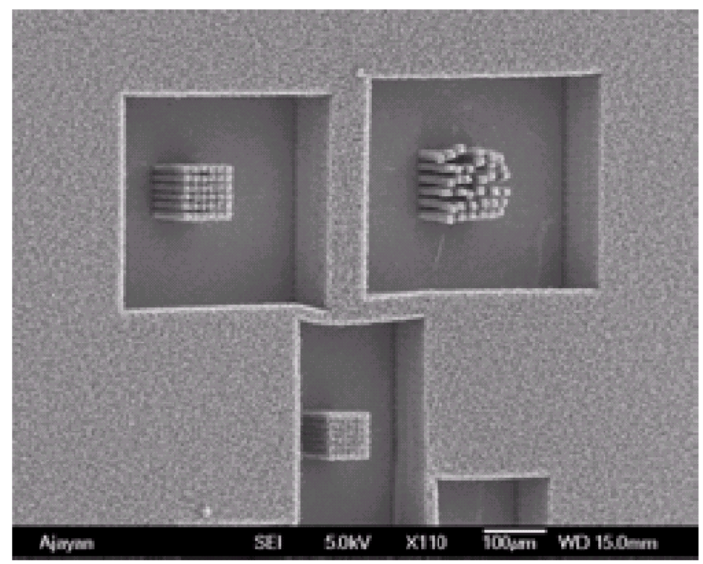
Nanotubes



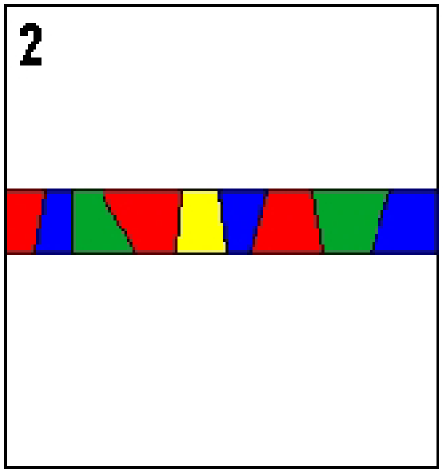




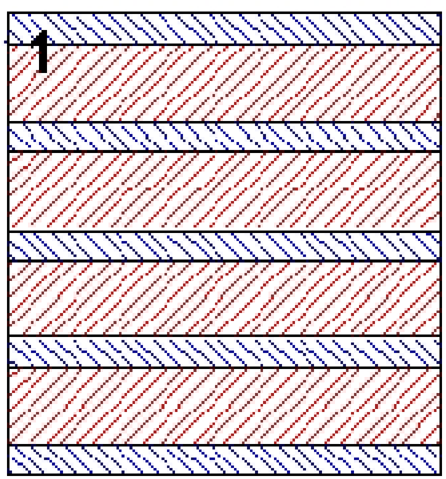




Nanofilm: any films with thickness in the nanometer range.

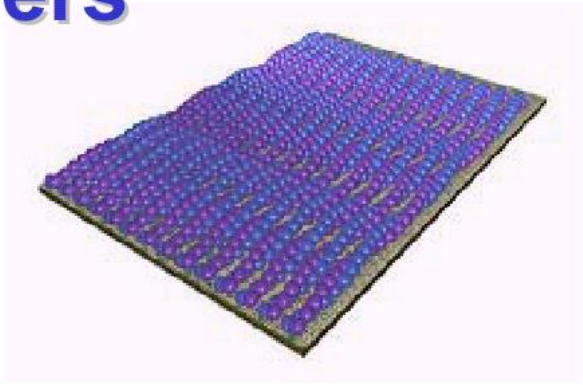


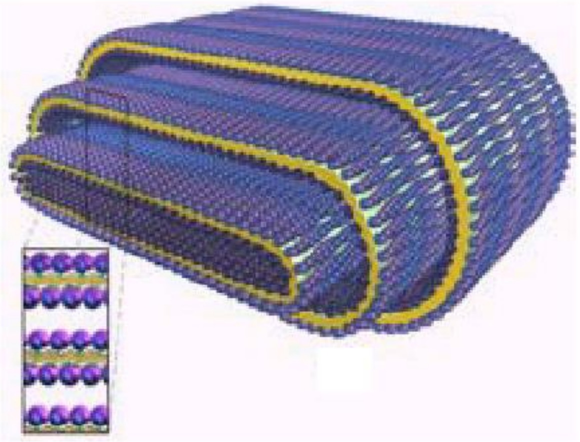
Single layer nanofilm: with a nanometer thickness; consisting of ultrafine grains (nanometer range diameter) (coatings, buried layers and thin films).

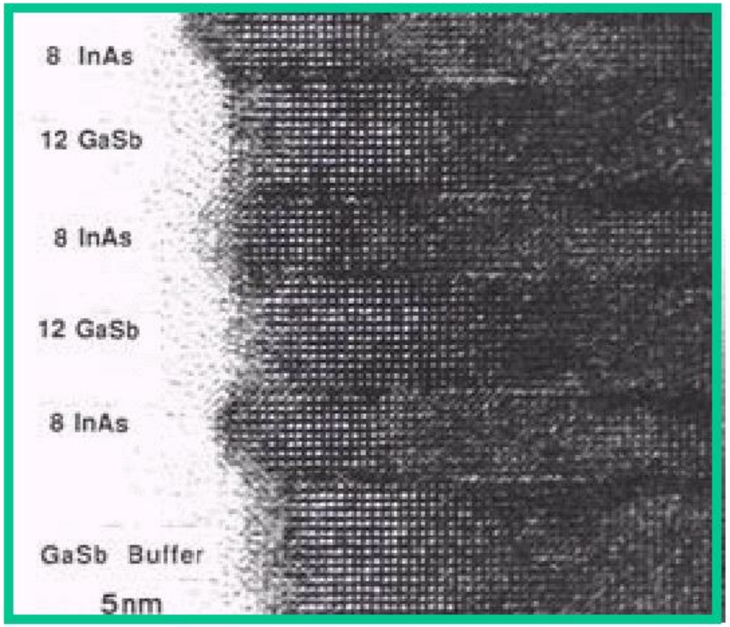


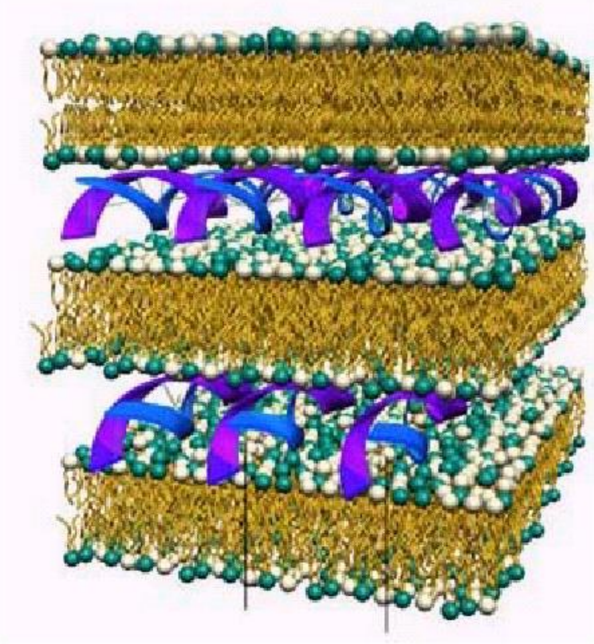
Multilayer nanofilm: with more than one layers with thicknesses in the nanometer range.

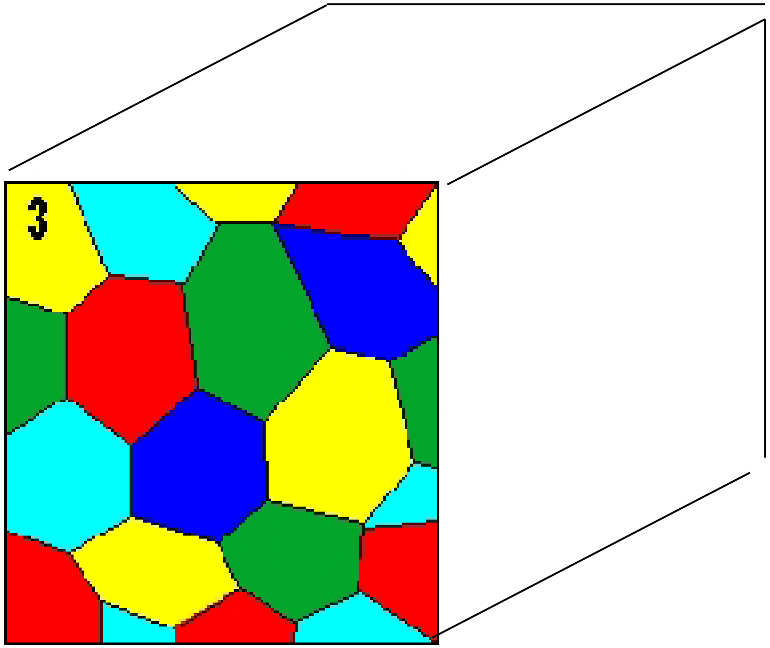
Nanolayers











Bulk nanomaterial: consisting of nanometer grains.

(synthesis and processing)

Bottom Up - from atom building block up to nanoscale material

Atomic or molecular precursors –colloid chemistry, gas-condensation, chemical precipitation, aerosol reactions, biological templating, sol-gel

Top down - from bulk material to nanoscale material

Bulk precursors - mechanical attrition, crystallization from the amorphous state, phase separation, plasma spray, electrochemical etching

Nucleation of supersaturated vapors

Issues in nanostructuring

- **Building blocks**
  - Size scale
  - Chemical composition
- **Assembly**
  - Interfaces and interactions
  - Modulation dimensionality
  - Architecture and hierarchy

构造和分级

- **Function**
  - Properties

Nanomaterials:

> size effect

> surface effect $\rightarrow$ unique properties

> quantum size effect

Nanoparticles exhibit size dependent properties that are profoundly different from the corresponding bulk material.

Electrical	higher electrical conductivity in ceramics and magnetic nanocomposites; higher resistivity in metals
Magnetic	increase in magnetic coercivity down to a critical size in the nanoscale regime; below critical crystalline size, decrease in the coersivity leading to the superparamagnetic behavior
Mechanical	increase in hardness and strength of metals and alloys; enhanced ductility, toughness and formability of ceramics; super strength and superplasticity
Optical	blue shift of optical spectra of quantum-confined crystallites; increase in luminescent efficiency of semiconductors
Chemical	fundamental understanding of heterogeneous catalytic properties

Magnetic coercivity: 磁矫顽力

Expected nanotechnology benefits

Medical applications

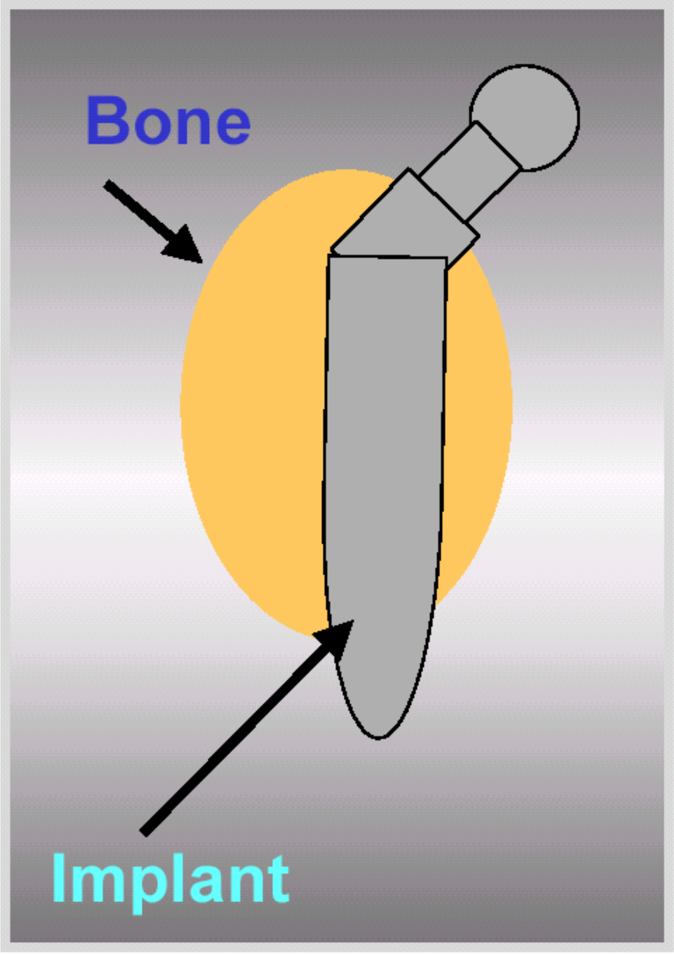
The medical potential of nanotechnologies is huge. Already on the market in the USA are **wound dressings** that exploit the antimicrobial (抗菌的) properties of nanocrystalline silver.

Nanoparticles such as nanocrystalline zirconium oxide (zirconia) and nanocrystalline silicon carbide are strong, lightweight, resistant to wear and corrosion and (unlike many other nanoparticles) inert.

For the future, **there is potential for nanoparticles to be used as vehicles for gene and drug delivery**.

Nanomaterials could make good **implants**.

Bone replacement



Implant requirements:

- mechanical behavior close to that of bone
- cellular compatibility
- osseointegration

Osseous: 骨的

(Webster, Siegel, Bizios 2000)

Environmental applications

There are environmental concerns about nanoparticles, but they could also play an important role in protecting the environment.

A typical application would be based on a column containing nanoparticles that bound to a particular contaminant.

As water passed through the column, the contaminant would be absorbed onto the nanoparticles.
The nanoparticles could then be retrieved (e.g. by removing them magnetically) and the contaminant washed out.

Military applications

New classes of nanopolymers are being developed that can be sprayed on to a soldier, to form a suit without seams (接缝). The fabric is planned to contain embedded enzymes that detect and break down chemical and biological warfare agents, various biosensors to monitor a soldier’s health, and nanosized silicon carbide particles for physical protection.

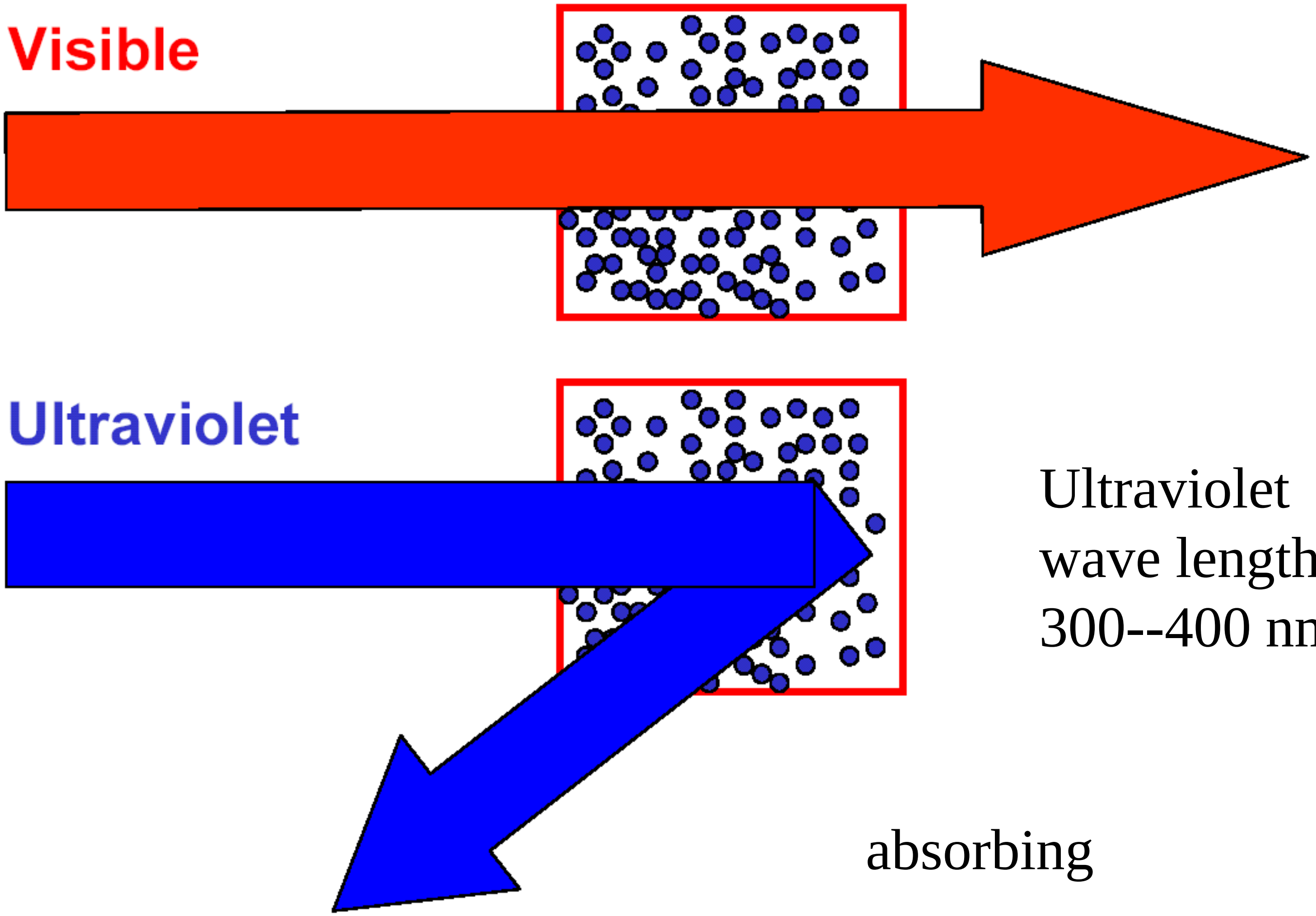
The nano-battle suit is being developed at Massachusetts Institute of Technology’s Institute for Soldier Nanotechnologies.

Cosmetics (化妆品)

Nano-titanium dioxide and zinc oxide can absorb and reflect UV light, while also being transparent to visible light. They are already used in sunscreens.

The cosmetics industry has invested heavily in nanotechnology. New products are claimed to penetrate deeper into the skin or to have other benefits. For example, cosmetics that slowly release vitamins are in development.

Protection from ultraviolet radiation

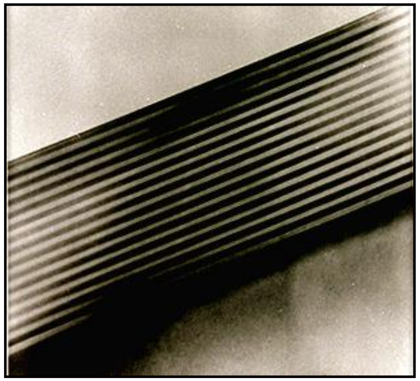


Ultraviolet
wave length:
300--400 nm.

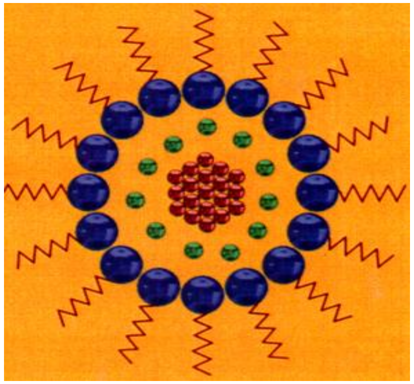
Materials with Enhanced Functionality via Nanostructuring

Nanoscience enables scientifically tailored materials

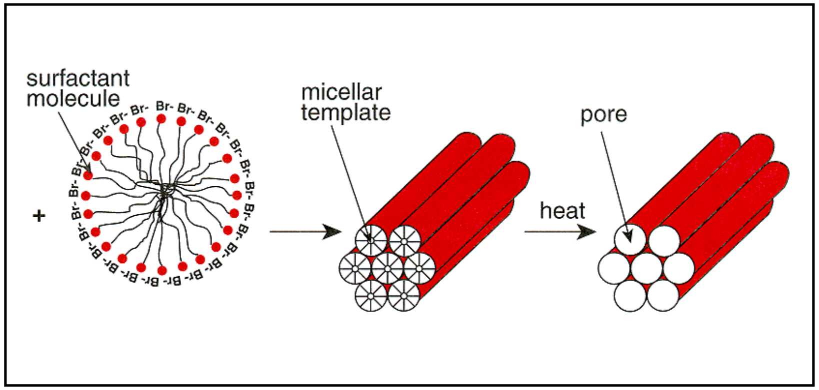
Layered-Structures



Nanocrystals



Nanocomposites



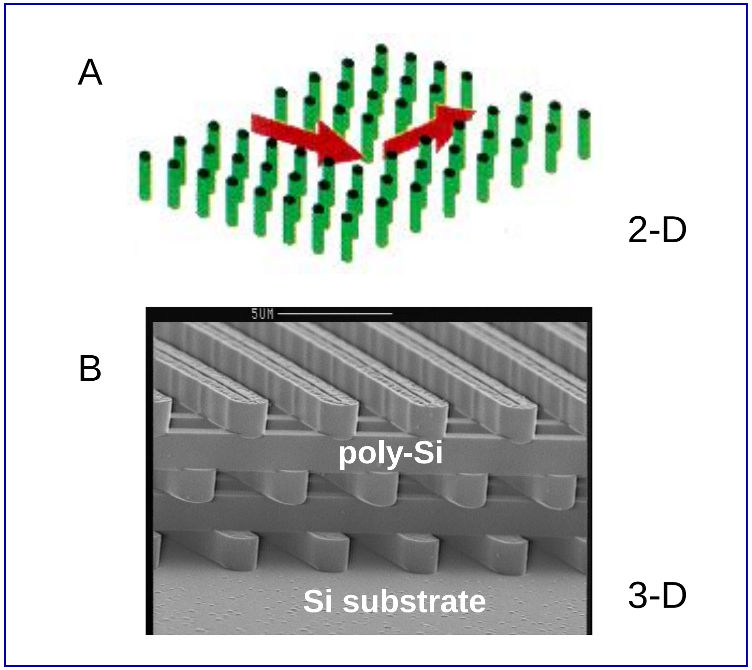
- **Electronics/photonics**
- **Novel Magnets**
- **Tailored hardness**

- **Catalysts**
- **Tailorable light emission**
- **Supercapacitors**

- **Separation membranes**
- **Adaptive/responsive behavior**
- **Pollutant/impurity gettering**

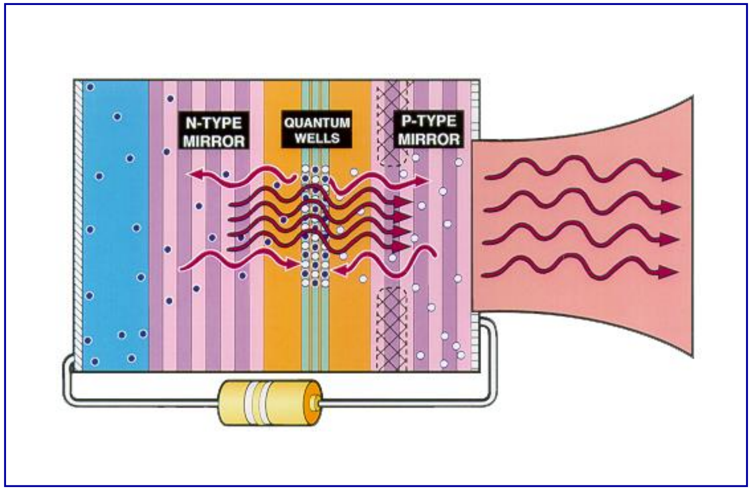
Materials with New Optical Properties via Nanostructuring

Photonic Lattices



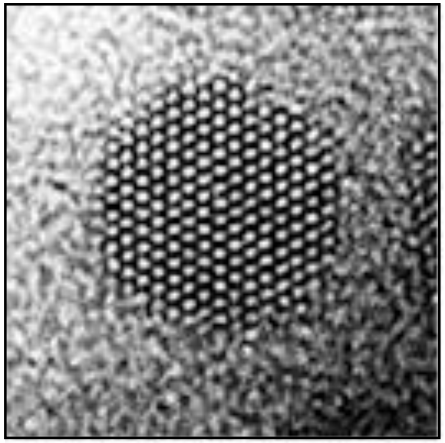
- Optical signals guided through narrow channels and around sharp corners
- Near 100% transmission
- Key technology for telecommunications and optical computing

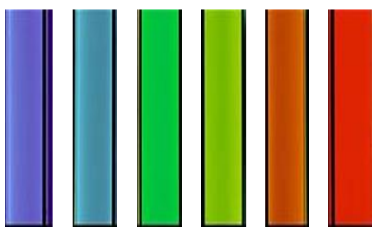
Vertical Cavity Surface Emitting Lasers (VCSELs)



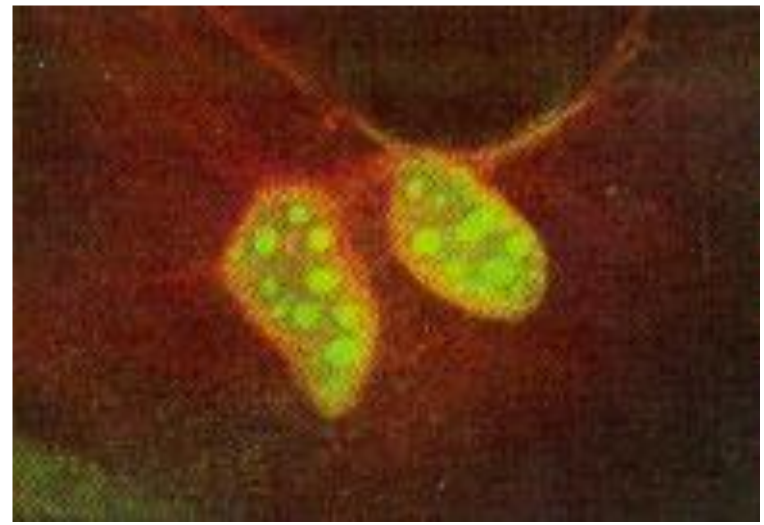
- The VCSEL is to photonics what the transistor was to electronics. A key 21st century technology
- Most efficient, low-power light source (57% in ‘97)
- Applications in stockpile stewardship, optical communications, scanners, laser printing, computing...

The Promise of Addressing Old Problems in New Ways





发荧光



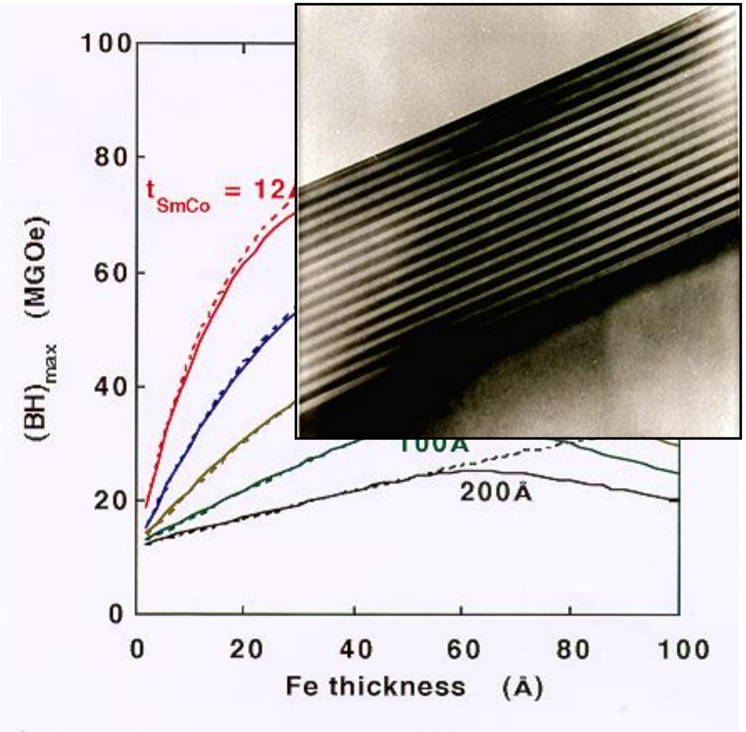
- Nanocrystals of CdSe fluoresce with different colors depending only on their size
- Different sized crystals can be selectively bound to different parts of a cell or to any desired structure to “light up” the parts

Semiconductor nanocrystals linked to bio-molecules light-up a cell’s actin filaments (red) and nucleus (green)

- Biological labeling
- Molecular processes in cells

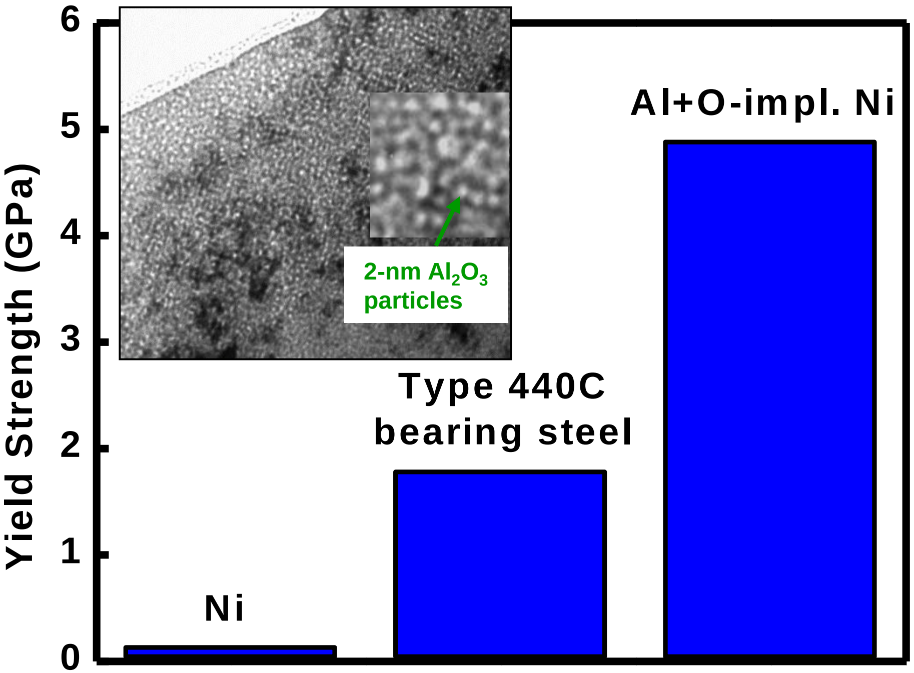
Materials for Improved Energy Efficiency and Performance

Exchange-Spring Magnets
SmCo/Fe

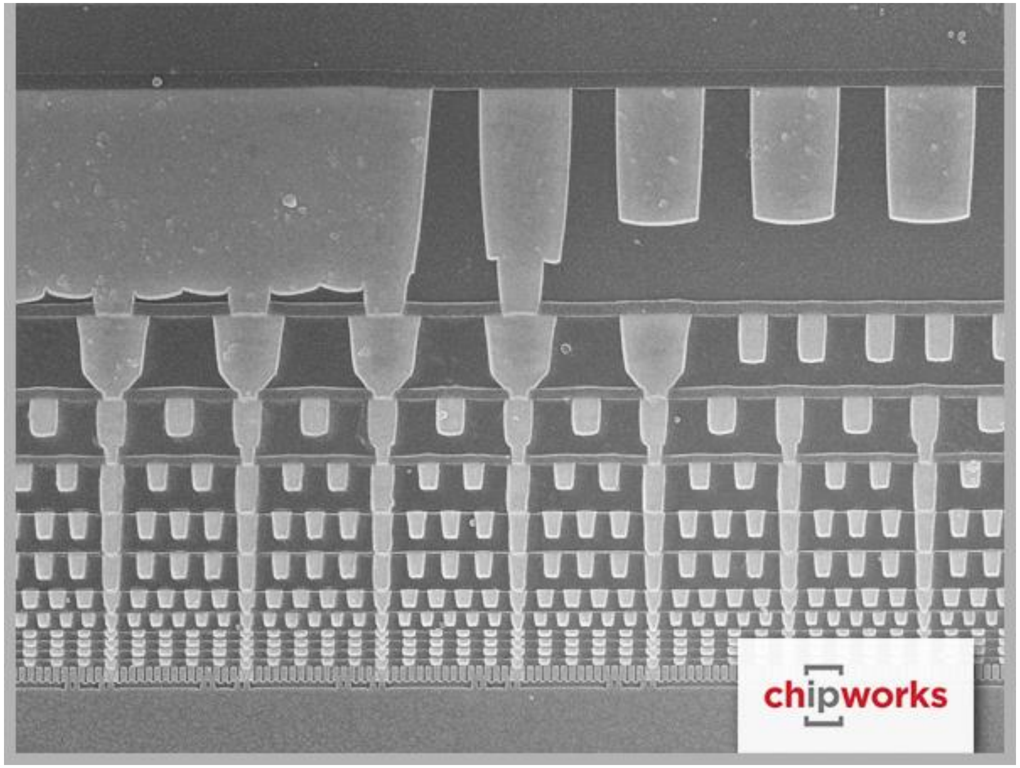


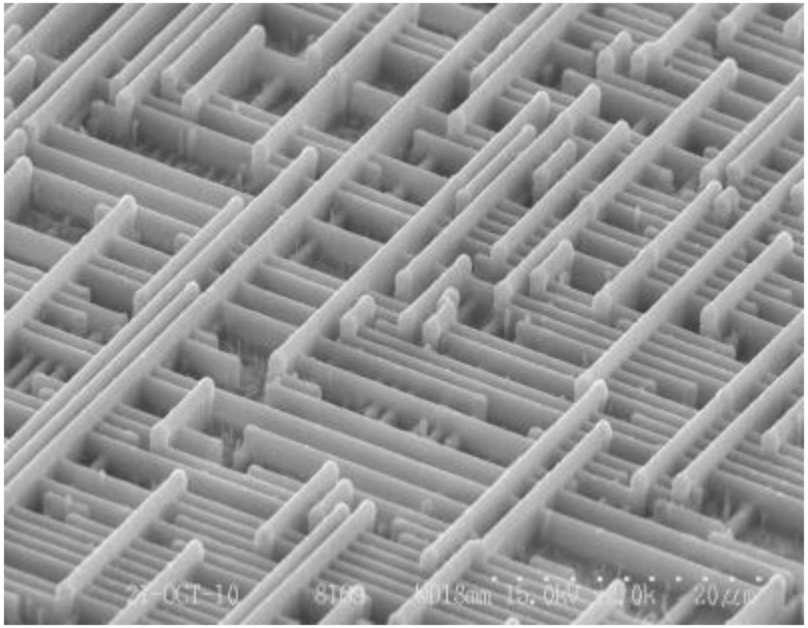
- Tailorable magnetic properties
- Lighter, stronger magnets
- More efficient motors

Ion-Implantation Metallurgy
Al+O Implanted Ni



- Superior strength
- Hard thin layers
- Greatly reduced friction & wear





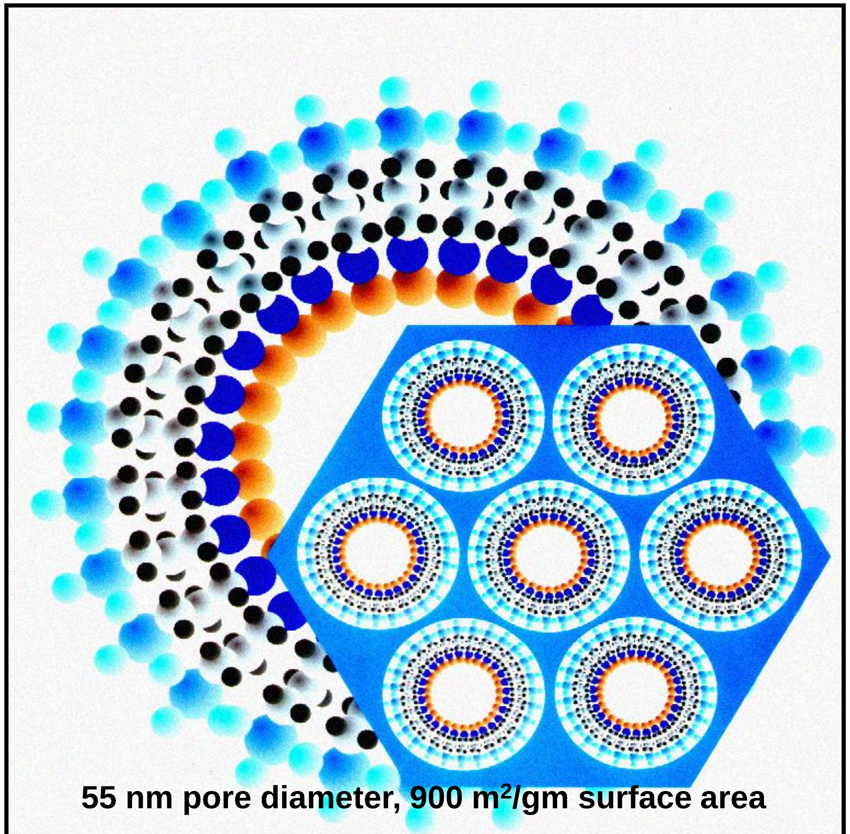
Wire crossing in the chip

Intel's 14nm Broadwell chip

3-D Self-Assembled Materials via Nanostructuring

单分子层

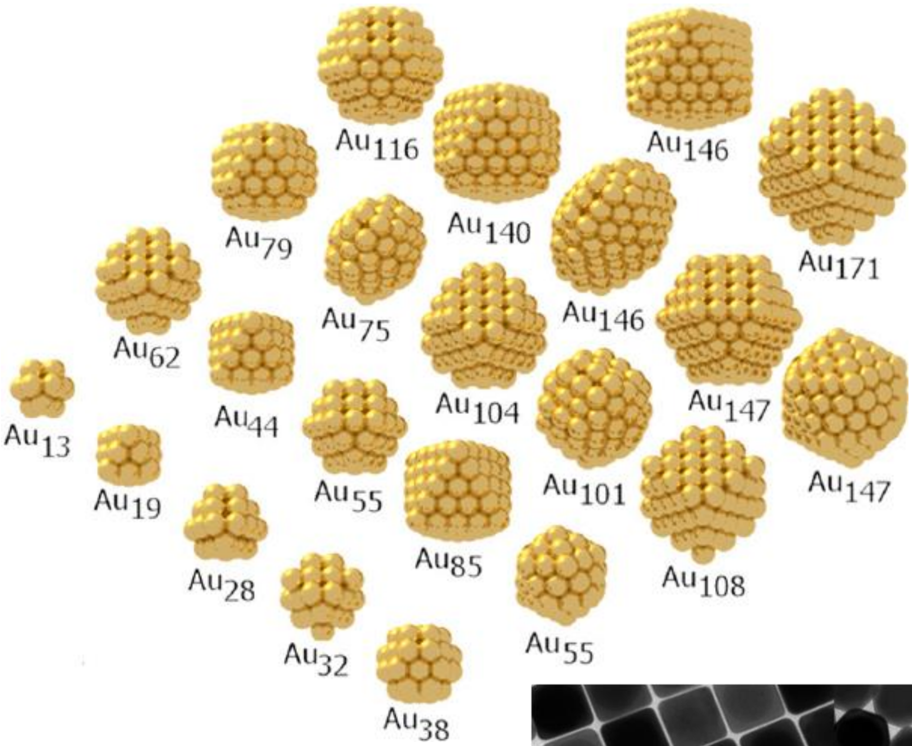
Self-Assembled Monolayers on Mesoporous Supports

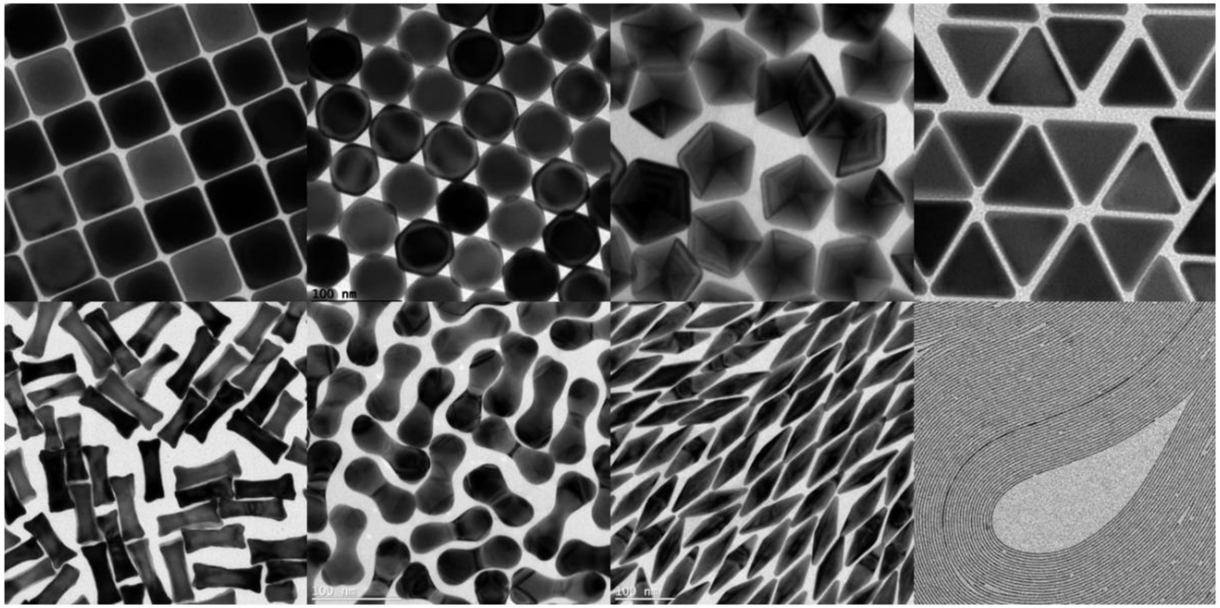


55 nm pore diameter, 900 m^2/gm surface area

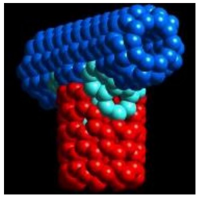
- Chemically selective surfactant molecules self-assemble within the interstices of a mesoporous silica matrix derived through solution processing routes.
- Resulting material shows high adsorption capacity for mercury and other heavy metals.
- Numerous environmental and commercial applications.

Self-Assembled Monolayers of Au Nanoparticles

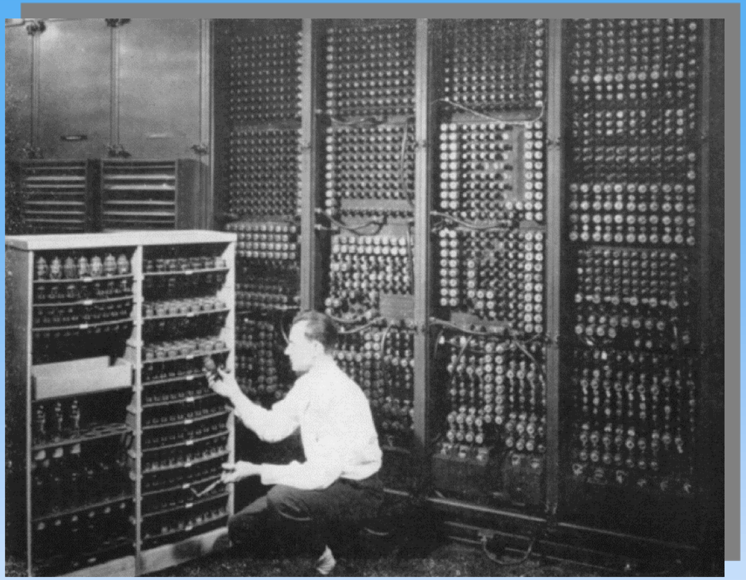




Nanoelectronics and Computing

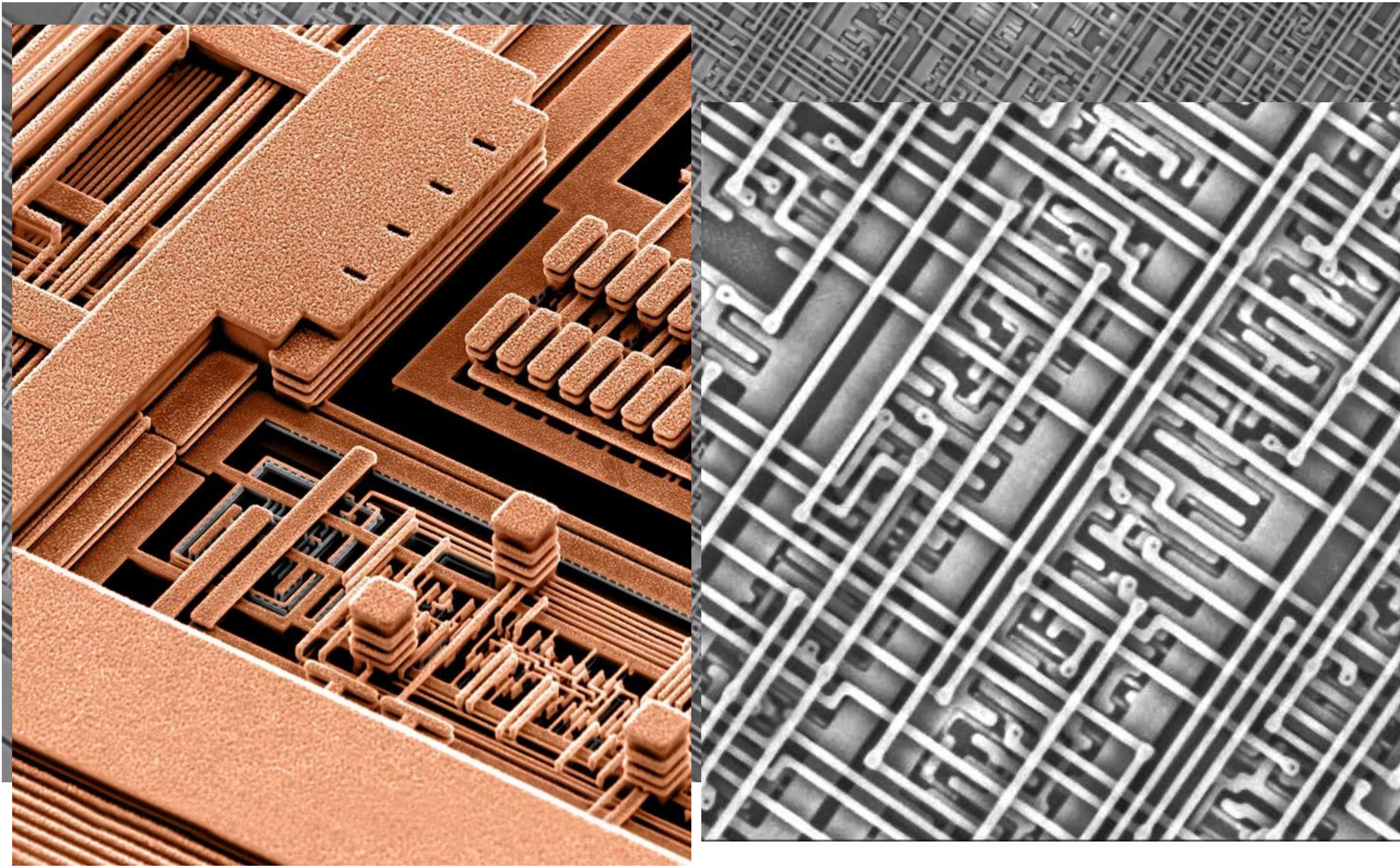


Past
Shared computing → thousands of people sharing a mainframe computer

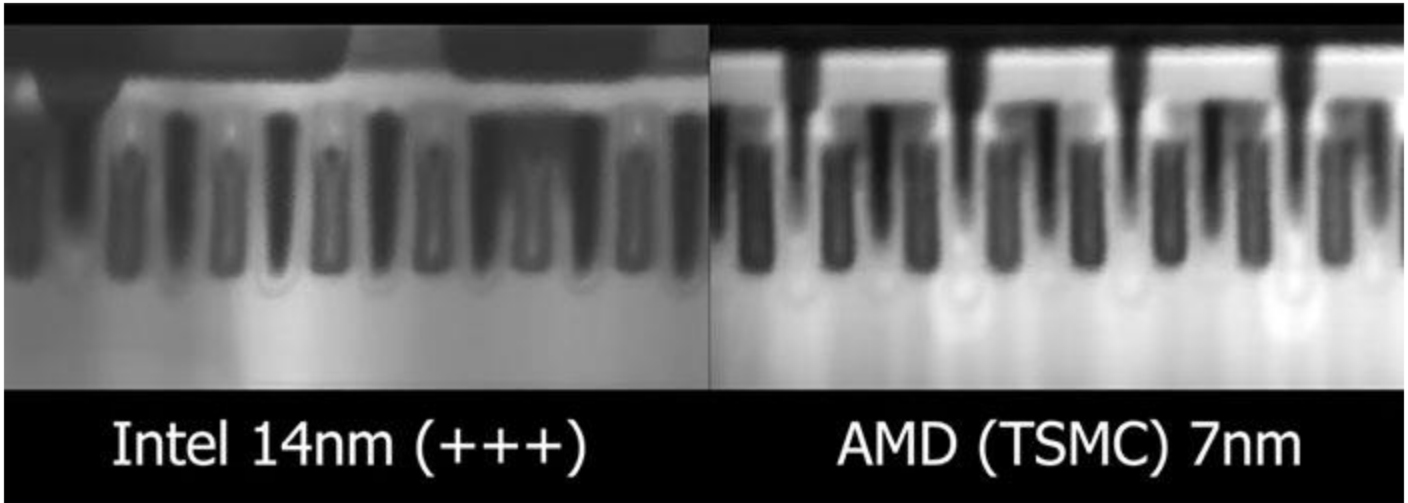


Present
Personal computing

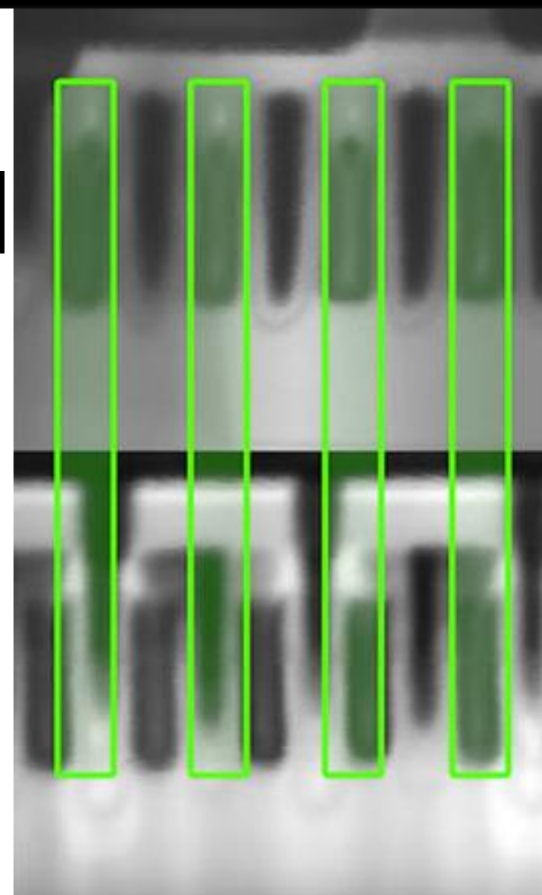
Future
Ubiquitous computing → thousands of computers sharing each and everyone of us; computers embedded in walls, chairs, clothing, light switches, cars….; characterized by the connection of things in the world with computation.



All of these pathways lead to transistors,



Intel 14nm (+++)



AMD (TSMC) 7nm

Intel 14nm and AMD/TSMC 7nm transistors micro-compared

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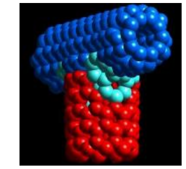
Nanoelectronics and Computing
Some Recent Advances



- Quantum Computing
  - Takes advantage of quantum mechanics instead of being limited by it
  - Digital bit stores info. in the form of ‘0’ and ‘1’; qubit may be in a superposition state of ‘0’ and ‘1’ representing both values *simultaneously* until a measurement is made
  - A sequence of N digital bits can represent *one* number between 0 and 2^N-1; N qubits can represent *all* 2^N numbers simultaneously

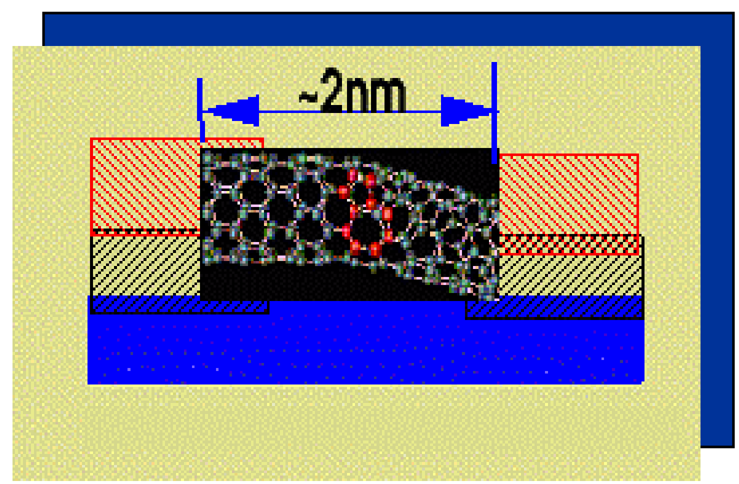
1938	1998
Technology engine: Vacuum tube	**Technology engine: CMOS FET**
Proposed improvement: Solid state switch	**Proposed improvement: Quantum state switch**
Fundamental research: Materials purity	**Fundamental research: Materials size/shape**

\- Stan Williams

- Carbon nanotube transistor by IBM and Delft University

- Molecular electronics: Fabrication of logic gates from molecular switches using rotaxane molecules

- Defect tolerant architecture, TERAMAC computer by HP → architectural solution to the problem of defects in future molecular electronics



Letter | Published: 25 September 2013

Carbon nanotube computer

Max M. Shulaker, Gage Hills, Nishant Patil, Hai Wei, Hong-Yu Chen, H.-S. Philip Wong & Subhasish Mitra

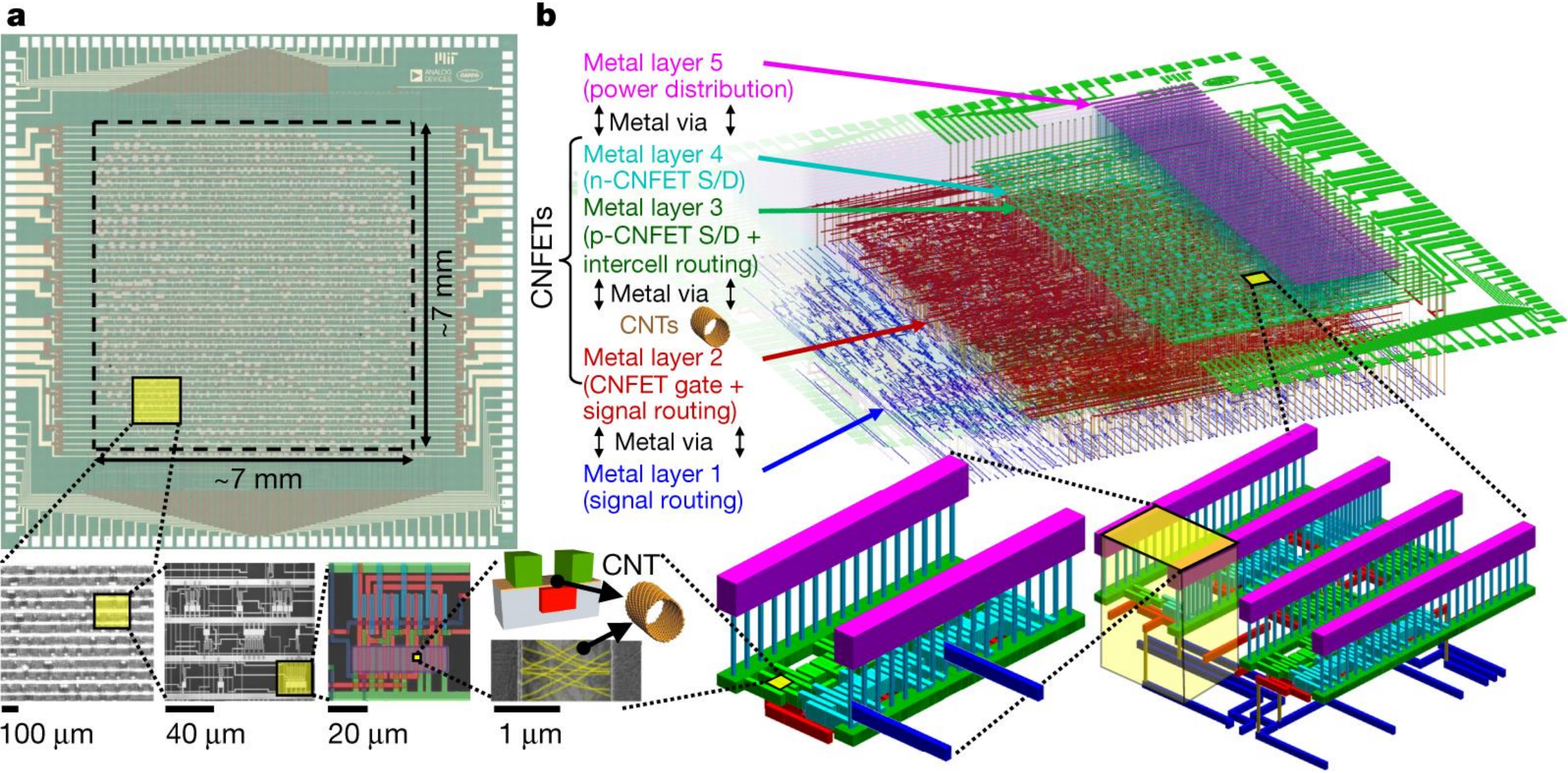
Nature **501**, 526–530(2013) | Cite this article

Article | Published: 28 August 2019

Modern microprocessor built from complementary carbon nanotube transistors

Gage Hills, Christian Lau, Andrew Wright, Samuel Fuller, Mindy D. Bishop, Tathagata Srimani, Pritpal Kanhaiya, Rebecca Ho, Aya Amer, Yosi Stein, Denis Murphy, Arvind, Anantha Chandrakasan & Max M. Shulaker

Nature **572**, 595–602(2019) | Cite this article



Nature **572**, 595–602(2019)

The MIT researchers have invented new techniques to dramatically limit defects and enable full functional control in fabricating CNFETs, using processes in traditional silicon chip foundries. They demonstrated a 16-bit microprocessor with more than 14,000 CNFETs that performs the same tasks as commercial microprocessors.

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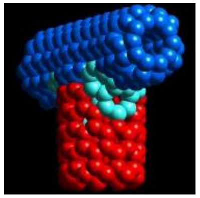
Expected Nanotechnology Benefits in Electronics and Computing



- Processors with declining energy use and cost per gate, thus increasing efficiency of computer by 10^6
- Higher transmission frequencies and more efficient utilization of optical spectrum to provide at least 10 times the bandwidth now
- Small mass storage devices: multi-tera bit levels
- Integrated nanosensors: collecting, processing and communicating massive amounts of data with minimal size, weight, and power consumption
- Quantum computing
- Display technologies

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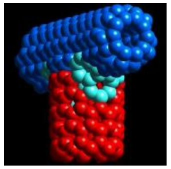
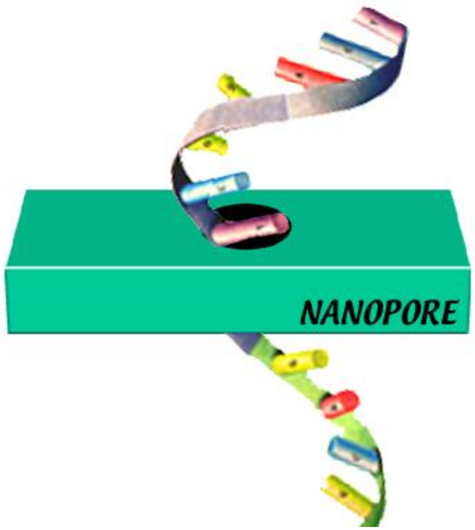
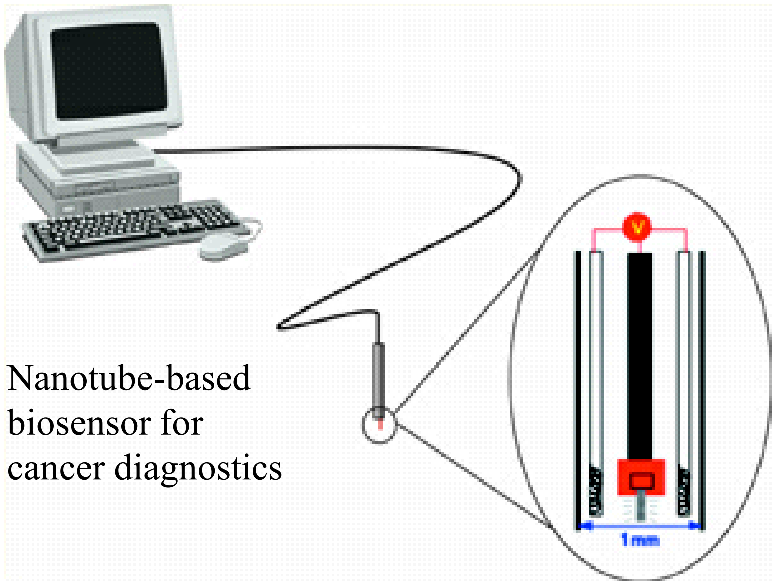
Health and Medicine





- Expanding ability to characterize genetic makeup will revolutionize the specificity of diagnostics and therapeutics
  - Nanodevices can make gene sequencing more efficient

- Effective and less expensive health care using remote and in-vivo devices



Nanotube-based biosensor for cancer diagnostics

- New formulations and routes for drug delivery, optimal drug usage

- More durable, rejection-resistant artificial tissues and organs

- Sensors for early detection and prevention

Health and Medicine
Some Recent Advances

- DNA microchip arrays using advances for IC industry
- 'Gene gun' that uses nanoparticles to deliver genetic material to target cells
- Semiconductor nanocrystals as fluorescent biological labels

Source: IWGN Report

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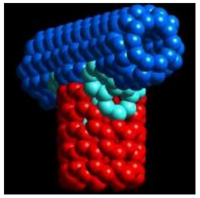
Benefits of Nanotechnology in Transportation



- Thermal barrier and wear resistant coatings
- High strength, light weight composites for increasing fuel efficiency
- High temperature sensors for ‘under the hood’
- Improved displays
- Battery technology
- Automatic highways
- Wear-resistant tires

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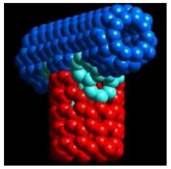
Energy and Environment



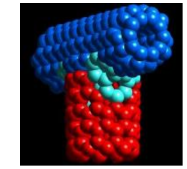
- Nanotechnology has the potential to impact energy efficiency, storage and production

- Materials of construction sensing changing conditions and in response altering their inner structure

- Monitoring and remediation of environmental problems; curbing emissions; development of environmental friendly processing technologies

- Some recent examples:
  - Crystalline materials as catalyst support, $300 b/year
  - Ordered mesoporous material by Mobil oil to remove ultrafine contaminants
  - Nano-particle reinforced polymers to replace metals in automobiles to reduce gasoline consumption

National Security



Some critical defense applications of nanotechnology include

- Continued information dominance: collection, transmission, and protection
- High performance, high strength, light weight military platforms while reducing failure rates and life cycle costs
- Chemical/biological/nuclear sensors; homeland protection
- Nano and micromechanical devices for control of nuclear and other defense systems
- Virtual reality systems based on nanoelectronics for effective training
- Increased use of automation and robotics

How Nanotechnology Can Change Your Life (video)